

Ultimate Phase Limits:

Coherence Horizons and Stability Bounds

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January 9, 2026

Subject Areas: Foundations of Physics / Unified Theory

Keywords: phase admissibility, coherence horizon, stability bounds, information density limits, phase curvature, emergence, Phase Theory

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Abstract

Phase Theory proposes that phase Φ is the sole fundamental constituent of physical reality, with all observable structures, fields, particles, and interactions arising as admissible configurations of the relational phase substrate. In this paper, Paper 15 of the Phase Theory series, we derive the ultimate physical limits imposed by Phase Theory by identifying absolute bounds on coherence, distinguishability, information density, curvature, and structural stability. These limits arise not from empirical constants inserted by hand, but from global phase admissibility constraints intrinsic to the phase substrate Φ . We rigorously define coherence horizons — the maximal relational extents over which phase coherence can be maintained under admissibility — and establish maximal phase density and curvature bounds from first principles, deriving them from the structure of the admissibility functional $I[\Phi]$ without appeal to external axioms.

We prove that physical reality neither diverges nor collapses because the admissibility functional $I[\Phi]$ simultaneously forbids both extremes, confining all physical

configurations to a narrow but robust stability regime $E \subset P \subset \{\Phi : I[\Phi] \geq 0\}$. The anti-fragmentation theorem and the anti-freezing theorem together show that admissible phase configurations are simultaneously stable against dissolution into incoherent microstructure and against stasis. We derive bounds on computation, observation precision, temporal variation of constants, and information storage capacity as consequences of phase admissibility, and demonstrate that each bound reduces to a known result — the Margolus-Levitin theorem, the Bekenstein bound, the Heisenberg uncertainty principle, and the Holevo bound — in appropriate limiting regimes, while providing Phase Theory corrections in high-curvature regimes.

The results explain why nature exhibits stable laws, finite complexity, and bounded information capacity without invoking fundamental discreteness, Planck-scale axioms, or anthropic selection. We additionally derive slow but nonzero drift rates for all emergent physical constants and specify correlated observable signatures. Eight classes of falsifiable predictions are enumerated and specified to the level of experimental protocol, discriminating signal size, and comparison against standard theoretical expectations. We provide systematic comparison with entropic bounds, holographic principles, thermodynamic limits, loop quantum gravity, string theory, and causal set approaches throughout. Four appendices provide mathematical conventions, derivation sketches, thermodynamic comparisons, and numerical estimates in SI units.

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§1. Why Ultimate Limits Must Exist

Every consistent physical theory must possess ultimate limits. This is not a philosophical preference but a mathematical necessity: in the absence of absolute bounds on energy, information, curvature, or coherence, the dynamical equations of any theory admit divergent solutions, self-referential inconsistencies, and infinite proliferation of degrees of freedom. The classical example is the ultraviolet catastrophe of pre-quantum electrodynamics, wherein the equipartition theorem, applied without restriction, distributed energy into arbitrarily short-wavelength modes and predicted infinite total radiated power. The resolution — quantum mechanics and the discretization of action by Planck's constant — introduced a limit not derived from classical principles but *imposed* by empirical observation and later rationalized theoretically. This pattern of external imposition is, we argue, the central unsatisfactory feature of modern theoretical physics.

Consider the catalogue of limits in the current theoretical framework. The Planck length $\ell_{Pl} = (\hbar G/c^3)^{1/2} \approx 1.616 \times 10^{-35} \text{ m}$ is posited as the minimum meaningful length, but no derivation from first principles within general relativity or quantum field theory establishes why this length is special; it is assembled dimensionally from three independently measured constants. The Planck energy $E_{Pl} = (\hbar c^5/G)^{1/2} \approx 1.956 \times 10^9 \text{ J}$ is similarly constructed. The Bekenstein entropy bound $S \leq 2\pi RE/\hbar c$ [Bekenstein, 1973] is derived from combining thermodynamics with general relativity and the generalized second law, but its derivation requires assuming the second law holds in curved spacetime, which is itself unproven in quantum gravity. The Margolus-Levitin theorem [Margolus & Levitin, 1998] bounds computational

speed by energy, but treats energy as a primitive. These limits are real and important; our complaint is not their existence but their *ad hoc provenance*.

Phase Theory takes a fundamentally different approach. Its central claim, established in Paper 1 [Hanks, 2026a], is that phase Φ is the unique ontological primitive and that all physical structures are configurations of Φ over the relational substrate Ω . Given this, the question of physical limits becomes: what configurations of Φ are physically realizable? The answer is provided by the admissibility functional $I[\Phi]$: a configuration is physically realizable if and only if $I[\Phi] \geq 0$. This single criterion, derived from the self-consistency of the phase description (Papers 1-2), generates all physical limits automatically.

To see why limits are necessary from the admissibility condition, consider what happens if no limit exists. Suppose phase curvature $\kappa[\Phi](x)$ is unbounded. Then the curvature penalty term in $I[\Phi]$ can be made arbitrarily negative at a point x . But $I[\Phi]$ is an integral over all of Ω , and if the integrand diverges negatively at any point faster than it can be compensated by the remainder of the domain, then $I[\Phi] \rightarrow -\infty$, rendering the configuration inadmissible. More precisely, suppose $\kappa[\Phi](x_0) > M$ for some large M . The penalty term contributes $-c_2 \kappa^2$ to the integrand. For $M \rightarrow \infty$, the integral over even a small neighborhood of x_0 diverges to $-\infty$, so $I[\Phi] \rightarrow -\infty$ regardless of the behavior elsewhere. Therefore, no configuration with unbounded curvature is admissible. The same argument applies to information density and phase gradient. This demonstrates mathematically that Phase Theory is self-bounding: the admissibility criterion *is* the limit.

Limit Type	Standard Physics Treatment	Phase Theory Treatment
Minimum length	Postulated: Planck length ℓ_{Pl} from dimensional analysis; not derived	Derived: measurement horizon μ_c from admissibility distinguishability (§39)
Maximum energy density	UV cutoff imposed: lattice, Pauli-Villars, or dimensional regularization; renormalization required	Derived: $\rho_{\text{I}}^{\text{max}}$ from information density bound (§14-15)

Limit Type	Standard Physics Treatment	Phase Theory Treatment
Maximum curvature	Not bounded in classical GR; singularities allowed by Penrose-Hawking theorems	Derived: $\kappa_{\max}$ from admissibility curvature penalty (§11)
Minimum coherence	Decoherence treated as environmental: no universal bound; depends on system	Derived: ε_c from global admissibility (§6)
Computational speed	Margolus-Levitin: bound by energy; energy treated as primitive	Derived: bound by phase propagation speed and information density (§32)
Information storage	Bekenstein-Hawking: empirical combination of GR + thermodynamics	Derived: phase holographic bound from admissibility functional (§17)
Constant variation rate	Unconstrained in GR; Brans-Dicke adds kinetic term but no bound	Derived: bounded by admissibility and coherence horizon (§40)
Singularities	Inevitable in GR (Penrose 1965); avoided only by unproven quantum effects	Forbidden: Singularity Exclusion Theorem from $\kappa_{\max}$ (§12)

To make the argument fully explicit, consider the formal statement. Let $I[\Phi] = \int_{\Omega} L_{\text{admiss}}(\Phi, \partial\Phi, \partial^2\Phi) d^n\Omega$. Suppose no bound exists on the gradient $|\nabla\Phi|$. Then one can construct a sequence of configurations Φ_k with $|\nabla\Phi_k|_{\max} = k$. The gradient penalty term in L_{admiss} contains $-\alpha|\nabla\Phi|^2$ with $\alpha > 0$. Even if the phase itself remains bounded and topologically trivial, $I[\Phi_k] \leq C - \alpha k^2 \cdot \text{vol}(\Omega_{\text{peak}}) \rightarrow -\infty$ as $k \rightarrow \infty$. Hence every unbounded-gradient configuration is inadmissible. The conclusion is inescapable: admissibility is a bounding principle. Every physical limit studied in this paper is a specialization of this fundamental observation.

In subsequent sections, we make each limit precise. The remainder of the paper is organized as follows: §2–§4 provide the mathematical foundations of admissibility. §5–§10 develop coherence horizons in full generality. §11–§17 establish curvature and information density bounds. §18–§25 prove the stability theorems. §26–§30 derive the universal constants. §31–§35 bound computation. §36–§39 bound

observation. §40–§41 address constant drift. §42–§55 develop implications, comparisons, predictions, and conclusions.

§2. Phase Admissibility as a Bounding Principle

The concept of admissibility was introduced in Paper 1 [Hanks, 2026a] and refined in Paper 2 [Hanks, 2026b]. We review and extend the essential framework here. The phase field Φ is defined on the relational substrate Ω , a measure space encoding all relational degrees of freedom prior to the emergence of spacetime. A configuration $\Phi: \Omega \rightarrow S^1$ is *physically realizable* if and only if it is admissible in the sense of the following condition.

Definition 2.1 (Admissibility).

A phase configuration $\Phi \in C^2(\Omega, S^1)$ is *admissible* if and only if the admissibility functional satisfies

$$(15.1) \quad I[\Phi] = \int_{\Omega} L_{admiss}(\Phi, \nabla\Phi, \nabla^2\Phi) d^n\Omega \geq 0,$$

where $d^n\Omega$ is the relational measure on Ω and L_{admiss} is the admissibility Lagrangian density defined in §3.

The fundamental significance of admissibility is that it is not an additional constraint imposed on top of a pre-existing theory, but rather the *definition of physical reality* within Phase Theory. There is no concept of a "physically realizable but inadmissible" configuration; the two are definitionally equivalent by Paper 1 Axiom A3. This is the pivotal move that makes Phase Theory self-bounding: because admissibility is constitutive of physical reality, and because $I[\Phi]$ contains penalty terms that become negative for extreme configurations, extreme configurations are not merely improbable or energetically disfavored — they are *not physical* in any sense.

The structure of L_{admiss} is motivated in Appendix B and derived in detail in Paper 1. Its three primary components are:

- **Gradient penalty:** $-\alpha/|\nabla\Phi|^2$, which penalizes rapid spatial variation of phase. The coefficient $\alpha > 0$ sets the scale of the penalty.
- **Curvature penalty:** $-\beta\kappa(\Phi)$, which penalizes configurations with high phase curvature. Here $\kappa(\Phi) = |\nabla^2\Phi|^2$ and $\beta > 0$.
- **Coherence reward:** $+\gamma C(\Phi)$, which rewards configurations with high phase coherence. Here $C(\Phi) = |\int e^{i\Phi} d\Omega|^2/|\Omega|^2$ and $\gamma > 0$.

From this structure, we can immediately deduce the sense in which admissibility is a bounding principle. The penalty terms make $I[\Phi]$ increasingly negative as $|\nabla\Phi|$ or $\kappa(\Phi)$ grow without bound, while the coherence reward is bounded above by γ (since $C(\Phi) \leq 1$). Therefore, for any sufficiently extreme configuration, $I[\Phi] < 0$, and the configuration is inadmissible.

We now establish the existence and uniqueness of the critical admissibility threshold.

Proposition 2.1 (Existence and Uniqueness of Critical Admissibility Threshold).

There exists a unique critical value $I_c = 0$ such that configurations with $I[\Phi] \geq I_c$ are admissible and those with $I[\Phi] < I_c$ are inadmissible. The value $I_c = 0$ is uniquely determined by the requirement of global phase consistency (Paper 1, Axiom A3).

Proof Sketch.: ○○○○○○○○

The critical threshold at zero has deep significance: it means that the boundary of physical reality is the zero set of $I[\Phi]$, a hypersurface in the infinite-dimensional configuration space. Configurations that approach this boundary from above (i.e., with $I[\Phi] \rightarrow 0^+$) represent the most extreme physical states: near-maximal curvature, near-minimal coherence, or near-maximal information density. These correspond to black hole horizons, the Big Bang, and maximum-entropy states, respectively. We will return to each of these correspondences in later sections.

A further consequence is that the space of admissible configurations, $P = \{\Phi : I[\Phi] \geq 0\}$, is a closed subset of $C^2(\Omega, S^1)$ in the topology induced by the C^2 norm. This closedness ensures that limits of admissible sequences are admissible, a technical requirement for well-posedness of the phase equation of motion. We derive the maximum admissible phase complexity per coherence region in §3 and §14 as a consequence of the functional structure established here.

§3. The Admissibility Functional: Mathematical Foundations

We now provide the complete mathematical specification of the admissibility functional $I[\Phi]$. This section constitutes the technical core from which all subsequent results are derived.

3.1 The Phase Field

Let $(\Omega, g, d^n\Omega)$ be the relational substrate: a smooth n -dimensional Riemannian manifold with metric g and measure $d^n\Omega = |\sqrt{\det g}|, d^n x$. In the most general formulation, Ω is a principal $U(1)$ bundle over a relational base manifold M , and the phase field is a section $\Phi \in \Gamma(P)$. For the purposes of this paper, we work with the simplest realization: $\Phi: \Omega \rightarrow S^1 \approx \mathbb{R}/(2\pi\mathbb{Z})$, i.e., a circle-valued scalar field. This corresponds to a single $U(1)$ degree of freedom; the generalization to higher gauge groups is discussed in Paper 5 [Hanks, 2026e].

The full admissibility Lagrangian density is:

$$(15.2) \quad L_{\text{admiss}}(\Phi, \nabla\Phi, \nabla^2\Phi) = \gamma C(\Phi) - \alpha |\nabla\Phi|^2 - \beta K(\Phi) + \lambda T(\Phi),$$

where the four terms are defined as follows:

- **Coherence term:** $C(\Phi) = |\int_V e^{i\Phi} d^n\Omega|^2 / |V|^2$, where V is a local coherence volume. This term is dimensionless and satisfies $0 \leq C(\Phi) \leq 1$ by the Cauchy-Schwarz inequality.
- **Gradient penalty:** $|\nabla\Phi|^2 = g^{\mu\nu}(\partial_\mu\Phi)(\partial_\nu\Phi)$, the squared magnitude of the phase gradient in the relational metric.

- **Curvature penalty:** $K(\Phi) = g^{\mu\nu} g^{\rho\sigma} (\nabla_\mu \nabla_\rho \Phi)(\nabla_\nu \nabla_\sigma \Phi) = |\nabla^2 \Phi|^2$, the squared Hessian norm.
- **Topological term:** $T(\Phi) = (1/2\pi) \int d\Phi$, the topological winding contribution, which is integer-valued for closed configurations. Its coefficient λ may be positive or negative depending on the topological sector (Paper 7 [Hanks, 2026g]).

The coefficients $(\alpha, \beta, \gamma, \lambda)$ are the parameters of the admissibility functional. In the context of Phase Theory, these are not freely adjustable: they are constrained by the internal consistency requirements derived in Paper 1 and by the requirement that the emergent spacetime and constants agree with observation (Papers 3–5). Their numerical values in SI units are estimated in Appendix D.

3.2 The Euler-Lagrange Equations

Variation of $I[\Phi]$ with respect to Φ gives the Euler-Lagrange equation for admissible configurations:

$$(15.3) \quad \delta I / \delta \Phi = \gamma \delta C / \delta \Phi + 2\alpha \nabla^2 \Phi + 2\beta \nabla^4 \Phi + \lambda \delta T / \delta \Phi = 0.$$

Equation (15.3) is the *phase admissibility equation*. Its solutions constitute the set of stationary points of $I[\Phi]$, which include both maxima (maximally admissible configurations, corresponding to physical states) and saddle points (unstable marginal configurations). We note that (15.3) is a fourth-order partial differential equation in Φ , owing to the curvature penalty term $\beta \nabla^4 \Phi$. This higher-order structure is essential: it provides a natural regularization of the phase dynamics at short scales and is responsible for the measurement horizon (§39) and the minimum length (§12).

3.3 The Phase Coherence Measure

Definition 3.1 (Phase Coherence Measure).

For a phase configuration Φ on a region $V \subset \Omega$, the *phase coherence measure* is defined as

$$(15.4) \quad \chi[\Phi; V] = |\int_V e^{i\Phi(x)} d^n \Omega| / |V|,$$

where $|V| = \int_V d^n \Omega$ is the relational volume of V .

Proposition 3.1.

For all admissible configurations and all regions $V \subset \Omega$, $0 \leq \chi[\Phi; V] \leq 1$.

Proof.: ○○○○○

The coherence measure χ attains its maximum value of 1 for a perfectly coherent configuration $\Phi = \text{const}$ (or more generally for a plane wave $\Phi = k \cdot x + \varphi_0$ over a region small compared to the coherence horizon). It attains its minimum of 0 for a configuration whose phase is uniformly distributed over S^1 , such as a configuration with many topological defects. The admissibility criterion enforces a lower bound $\chi[\Phi; V] \geq \varepsilon_c > 0$ for all coherence volumes V , derived from the requirement that $I[\Phi] \geq 0$ (see §6).

Admissible configurations are formally defined as solutions of (15.3) satisfying the global boundary condition that $I[\Phi] \geq 0$ is maintained as a global integral constraint, not merely locally. This distinction between local and global admissibility is the subject of the next section.

§4. Global vs. Local Admissibility Constraints

A subtlety of the admissibility framework — one with profound physical consequences — is the distinction between local and global admissibility. A configuration may satisfy all pointwise conditions on Φ and its derivatives while failing the global integral condition $I[\Phi] \geq 0$; conversely, a globally admissible configuration may permit local regions of high curvature or low coherence, provided the global integral remains non-negative. This distinction mirrors the distinction in topology between local and global properties: a manifold may be locally flat (zero

curvature at every point) while being globally curved (non-zero total Gauss-Bonnet invariant).

Definition 4.1 (Local Admissibility).

A configuration Φ is *locally admissible* at $x \in \Omega$ if the integrand satisfies $L_{admiss}(\Phi(x), \nabla\Phi(x), \nabla^2\Phi(x)) \geq 0$ pointwise.

Definition 4.2 (Global Admissibility).

A configuration Φ is *globally admissible* if $I[\Phi] = \int_{\Omega} L_{admiss} d^n\Omega \geq 0$.

Lemma 4.1 (Local Admissibility Lemma).

Every locally admissible configuration satisfies the pointwise gradient bound $|\nabla\Phi(x)|^2 \leq (\gamma C_{max} + \lambda T_{max})/\alpha$ and the pointwise curvature bound $\kappa(\Phi)(x) \leq (\gamma C_{max} + \lambda T_{max})/\beta$, where $C_{max} = 1$ and T_{max} is the maximum topological contribution per unit volume.

Proof.: ○○○○○○

Theorem 4.1 (Global Admissibility Theorem).

A globally admissible configuration satisfies integral bounds that cannot be derived from local admissibility alone. Specifically, for any globally admissible Φ :

$$(15.5) \quad \int_{\Omega} |\nabla\Phi|^2 d^n\Omega \leq (\gamma/\alpha) |\Omega| + (\lambda/\alpha) N_{top},$$

where $N_{top} = \int_{\Omega} T(\Phi) d^n\Omega$ is the total topological charge. This integral bound is strictly stronger than the pointwise bound from Lemma 4.1 when $|\nabla\Phi|$ is

concentrated in subregions.

Proof.: ○○○○○○

The physical content of Theorem 4.1 is profound: it shows that Phase Theory has global constraints — conservation laws, not just local equations of motion. The global phase consistency enforcement mechanism from Papers 1-2 is precisely the mechanism that enforces the integral bound (15.5). It acts like a global Gauss's law for phase: the total "phase flux" through any closed relational region is bounded by the topological charge enclosed.

Topological defects arise precisely at the boundary between local and global admissibility. A configuration may be locally admissible everywhere except at isolated points or submanifolds where local admissibility fails — these are topological defects (Paper 7 [Hanks, 2026g]). The defect density is bounded by the global constraint: the total topological charge N_{top} is constrained by (15.5), and hence the defect density cannot grow without bound in any admissible configuration. This provides the phase-theoretic explanation for confinement (Paper 8 [Hanks, 2026h]): free quarks would correspond to unconfined topological defects, whose proliferation is forbidden by global admissibility.

§5. Coherence Horizons: Conceptual Framework

Before providing the precise mathematical definition of coherence horizons in §6, we develop the physical and conceptual motivation. The core idea is simple: phase correlations cannot extend indefinitely. At large separations, the phase at two points becomes uncorrelated because the admissibility functional penalizes configurations with strong long-range correlations that would require either very large phase gradients (which are penalized by the gradient term) or very high curvature (penalized by the curvature term). The maximum separation over which coherence can be maintained is the coherence horizon.

To see why coherence horizons must exist from first principles, consider a one-dimensional analog. Suppose $\Phi(x)$ is coherent over a length L , meaning $|\Phi(x) - \Phi(y)| \ll \pi$ for all $|x - y| \leq L$. Coherence implies that the phase varies smoothly over L , so the average gradient is approximately $\Delta\Phi/L$ where $\Delta\Phi$ is the total phase variation over L . The gradient penalty in $I[\Phi]$ contributes approximately $-\alpha (\Delta\Phi/L)^2 L = -\alpha (\Delta\Phi)^2 / L$ to the admissibility. As L increases with fixed $\Delta\Phi$, this contribution becomes less negative — coherence over large scales is not *directly* penalized. However, maintaining coherence over large scales requires that phase variations are suppressed, i.e., $\Delta\Phi$ remains small. Large $\Delta\Phi$ at large L would either require a large gradient (penalized) or a highly oscillatory path (penalized by curvature). The admissibility functional balances these pressures to produce a characteristic scale: the coherence horizon.

The coherence horizon has no analog in classical field theory, where correlations decay but do not have a sharp cutoff. In quantum field theory, correlation lengths are set by particle masses: $\xi = \hbar/(mc)$. Phase Theory generalizes this: coherence horizons play the role of correlation lengths, but they are dynamical and depend on the full phase configuration, not just a single mass parameter. Indeed, the mass-based correlation length emerges from the coherence horizon in the limit where the phase configuration is a stable localized mode (Paper 2 [Hanks, 2026b]).

The comparison with other notions of "horizon" in physics is instructive. The cosmological (Hubble) horizon is the distance light has traveled since the Big Bang; it limits causal influence but does not limit quantum correlations. The de Sitter horizon similarly limits causal access. The decoherence radius in quantum mechanics is the scale over which a quantum system loses phase coherence due to environmental interaction; it is closely related to but not identical with the Phase Theory coherence horizon, since Phase Theory's coherence horizon is intrinsic to the phase substrate, not environmental. The correlation length in condensed matter physics is the closest analog: a characteristic length set by the competition between ordering tendencies and thermal fluctuations. In Phase Theory, the admissibility functional plays the role of the free energy, and the coherence horizon is the Phase Theory correlation length.

Coherence horizons explain several features of physical reality that are otherwise postulated or left unexplained. *Finite interaction range*: forces are mediated by phase modes, and their range is bounded by the coherence horizon of the corresponding mode. *Causal structure*: the causal light cone emerges from the propagation speed of coherent phase modes (Paper 3 [Hanks, 2026c]), and the maximum range of any causal influence is bounded by the coherence horizon times the propagation speed. *Emergence of locality*: physics appears local because coherence — and therefore physical influence — is confined to coherence volumes. *Particle confinement*: quarks cannot escape their coherence volumes; the boundary of the coherence volume is the confinement scale (Paper 8 [Hanks, 2026h]).

§6. Definition and Derivation of Coherence Horizons

We now provide the formal definition and derive the coherence horizon from the admissibility functional.

Definition 15.1 (Coherence Horizon).

For an admissible phase configuration Φ , the *coherence horizon* $\ell_c[\Phi]$ is the supremum of lengths L such that the two-point phase correlation function

$$(15.6) \quad G(x, x') = \langle e^{i(\Phi(x) - \Phi(x'))} \rangle$$

satisfies $|G(x, x')| \geq \varepsilon_c$ for all x, x' with $d(x, x') = L$, where $\varepsilon_c > 0$ is the minimum admissible coherence threshold determined by the admissibility functional, and $d(x, x')$ is the relational distance in Ω . The average $\langle \cdot \rangle$ is taken over the admissible phase ensemble.

The threshold ε_c is not a free parameter: it is fixed by the requirement that $I[\Phi] \geq 0$. Specifically, if $|G(x, x')| < \varepsilon_c$ for nearby points, the coherence term $C(\Phi)$ in the admissibility Lagrangian falls below the level needed to compensate the gradient and curvature penalties, rendering $I[\Phi] < 0$. The value of ε_c is determined by the

balance condition $\gamma \varepsilon_c^2 = \alpha \|\nabla \Phi\|_{typ}^2$, where $\|\nabla \Phi\|_{typ}$ is the typical gradient in the coherence region.

We now derive the coherence horizon from the admissibility functional. The key step is to compute $G(x, x')$ for a generic admissible configuration. For a configuration with approximately uniform gradient $k = |\nabla \Phi|$ in the coherence region, we have $\Phi(x) - \Phi(x') \approx k \cdot (x - x')$, so

$$(15.7) \quad |G(x, x')| \approx |(e^{ik \cdot (x-x')})| = e^{-\sigma_k^2 |x-x'|^2/2},$$

where σ_k^2 is the variance of the gradient in the ensemble. This Gaussian decay is characteristic of configurations near the vacuum (low curvature); the coherence horizon is determined by the condition $e^{-\sigma_k^2 \ell_c^2/2} = \varepsilon_c$, giving

$$(15.8) \quad \ell_c = (-2 \ln \varepsilon_c)^{1/2} / \sigma_k.$$

For configurations with non-negligible curvature, the decay is faster (approximately exponential rather than Gaussian), and the coherence horizon is shorter. Specifically, for high-curvature configurations, the two-point function decays as $e^{-|x-x'|/\ell_c}$ with

$$(15.9) \quad \ell_c = (\kappa/\alpha)^{1/2},$$

where κ is the curvature coupling and α is the gradient penalty coefficient. This formula (15.9) is the key result relating the coherence horizon to the admissibility functional parameters.

Theorem 15.1 (Finite Coherence Horizon).

Every admissible phase configuration has a finite coherence horizon: $\ell_c[\Phi] < \infty$ for all $\Phi \in P$.

Proof.

Suppose for contradiction that $\ell_c[\Phi] = \infty$. Then $|G(x, x')| \geq \varepsilon_c > 0$ for all x, x' in Ω . By the ergodic theorem for the phase ensemble, this implies that the phase field has no

large-scale variation in the sense that $\|\nabla\Phi\|_{L^2(\Omega)}^2 = 0$. But this means $\Phi = \text{const}$ almost everywhere. A constant phase field carries no information (entropy = 0) and corresponds to the vacuum Φ_0 . Any non-trivial configuration with $\Phi \neq \text{const}$ must have non-zero gradient, which by (15.7)–(15.9) leads to exponential decay of $|G(x, x')|$ and hence a finite ℓ_c . Therefore, every non-trivial admissible configuration has a finite coherence horizon. The trivial case $\Phi = \text{const}$ has $\ell_c = \infty$ technically, but carries no physical content and is excluded from the class of "physically interesting" configurations. $\square$

Corollary 15.1.

The coherence horizon ℓ_c is bounded above: $\ell_c[\Phi] \leq \ell_c^{\max} = (\kappa/\alpha)^{1/2} (-2 \ln \varepsilon_c)^{1/2}$ for all non-trivial admissible configurations.

The existence and finiteness of the coherence horizon establishes the fundamental locality of Phase Theory: no physical influence, correlation, or information transfer can extend beyond ℓ_c . This is the phase-theoretic origin of locality and causality, derived from the admissibility functional rather than postulated as a spacetime axiom.

§7. Spatial Coherence Horizons

We now specialize the general coherence horizon to spatial configurations, deriving the spatial coherence horizon $\ell_c^{(s)}$ and its dependence on local phase properties.

Definition 15.2 (Spatial Coherence Horizon).

The *spatial coherence horizon* $\ell_c^{(s)}$ is the coherence horizon of Definition 15.1 restricted to the spatial submanifold of Ω , i.e., computed using the spatial two-point correlation function at equal emergent time.

The spatial coherence horizon depends on three local phase properties: (1) the local phase gradient density $\rho_\phi(x) = |\nabla\Phi(x)|^2$, (2) the topological charge density $n_{top}(x)$ (number of vortex/defect cores per unit volume), and (3) the admissibility curvature coupling κ .

Starting from (15.9), we refine the formula to account for these effects. In regions with non-zero topological charge density, the phase field necessarily has large gradient near the defect cores (since the phase winds by 2π around each core). The effective gradient density in a region containing n_{top} defects per unit volume is

$$(15.10) \rho_\phi^{eff} = \rho_\phi + (2\pi)^2 n_{top} / a_c^2,$$

where a_c is the defect core size. Substituting into the coherence horizon formula gives

$$(15.11) \ell_c^{(s)} = (\kappa/\alpha)^{1/2} (\rho_\phi^{eff})^{-1/4}.$$

This formula reveals an important scaling law:

Proposition 15.2 (Spatial Coherence Scaling).

The spatial coherence horizon scales as

$$(15.12) \ell_c^{(s)} \sim \rho_\phi^{-1/2}$$

in the limit of low topological charge density and uniform phase gradient density, where $\rho_\phi = |\nabla\Phi|^2$ is the local phase information density.

Proof.: ○○○○○○

The scaling law (15.12) connects to observable physics in two important ways. First, from Paper 8 [Hanks, 2026h] on phase-theoretic confinement, the spatial coherence horizon corresponds to the confinement scale $\ell_{conf} \sim 1/\Lambda_{QCD}$: the regime where the phase information density becomes so high that phase coherence cannot be maintained over distances greater than $\ell_c^{(s)}$, confining quarks within hadrons. Second, from Paper 3 [Hanks, 2026c] on emergent spacetime, the spatial coherence

horizon determines the effective "grid scale" of emergent space: spatial structures smaller than $\ell_c^{(s)}$ are genuine physical degrees of freedom, while apparent structures at sub-coherence scales are artifacts of description.

The upper bound on $\ell_c^{(s)}$ follows from the maximum admissible value of $I[\Phi]$ and the minimum nonzero phase gradient density:

$$(15.13) \ell_c^{(s)} \leq (\kappa/\alpha)^{1/2} (\rho_\Phi^{\min})^{-1/2} = \ell_c^{\max},$$

where $\rho_\Phi^{\min}$ is the minimum nonzero gradient density in the physical regime, corresponding to the cosmological configuration. This upper bound defines the cosmological coherence horizon (discussed in §10 and §48).

§8. Temporal Coherence Horizons

Phase Theory treats time as an emergent feature of phase evolution (Paper 3 [Hanks, 2026c]), not a fundamental parameter. Nevertheless, once an effective emergent time coordinate t is established by the dominant phase propagation mode, it is meaningful to define temporal coherence: the maximum duration over which phase correlations in time are maintained.

Definition 15.3 (Temporal Coherence Horizon).

The *temporal coherence horizon* τ_c is the supremum of time intervals T such that the temporal two-point correlation function $G(t, t') = \langle e^{i(\Phi(t) - \Phi(t'))} \rangle$ satisfies $|G(t, t')| \geq \varepsilon_c$ for $|t - t'| = T$.

The temporal coherence horizon is bounded above by the spatial coherence horizon and the phase propagation speed. To see this, note that in an admissible configuration, the phase evolves by propagating spatial coherent modes at speed v_ϕ . A temporal correlation at separation T requires a spatial correlation at separation $v_\phi T$; if this exceeds $\ell_c^{(s)}$, the temporal correlation is zero. Therefore:

Theorem 15.2 (Temporal Coherence Bound).

$$(15.14) \quad \tau_c \leq \mathcal{L}_c^{(s)} / V_\Phi.$$

Proof.: ○○○○○

The temporal coherence horizon has three major physical implications:

Arrow of time. Beyond τ_c , the phase configuration at time $t + T$ carries no memory of the configuration at time t : temporal decoherence is complete. This irreversibility provides the phase-theoretic arrow of time. The arrow is not imposed by boundary conditions (as in standard thermodynamics) but emerges necessarily from the temporal coherence bound: the phase equation of motion (15.3) is time-reversal symmetric, but the decay of temporal correlations beyond τ_c is not, because the coherence of the reference configuration cannot be restored by admissible evolution alone (Anti-Fragmentation Theorem, §18).

Quantum coherence times. In quantum mechanics, the coherence time of a quantum system is the timescale over which its density matrix remains off-diagonal. In Phase Theory, this corresponds to the temporal coherence horizon of the phase configuration encoding the quantum state. The quantum Zeno timescale (the time over which the quantum survival probability deviates quadratically from 1) is related to the inverse of the energy spread; in Phase Theory, this corresponds to the inverse phase frequency spread, which is $\tau_{Zeno} \sim \mathcal{L}_c^{(s)} / V_\Phi = \tau_c$. The coincidence is not accidental: both reflect the same admissibility-constrained coherence timescale.

Time-energy uncertainty. From the temporal coherence bound, an admissible phase process occurring over a time interval $\Delta t \leq \tau_c$ has a minimum energy spread $\Delta E \geq \hbar / \tau_c$. This is the time-energy uncertainty relation, derived here from temporal coherence rather than postulated as a quantum axiom. The derivation makes clear that it is a structural consequence of the admissibility functional, not an observation about measurement disturbance.

§9. Relational Coherence and Non-Locality

The relational substrate Ω is more general than spatial or temporal manifolds. It encodes all relational degrees of freedom, including those that do not correspond to spatial separation in emergent spacetime. Coherence horizons on Ω therefore have implications beyond the spatial and temporal cases: they describe the maximum extent of relational phase correlations, which can be non-local in the emergent spatial sense.

Definition 15.4 (Relational Coherence Horizon).

The *relational coherence horizon* $\ell_c^{(r)}$ is the coherence horizon of Definition 15.1 computed with respect to the full relational distance $d_\Omega(x, x')$ on Ω , which may differ from the emergent spatial distance.

The key insight is that two points x and x' in Ω may be relationally close (small $d_\Omega(x, x')$) while being spatially distant in the emergent sense. This occurs when the relational degrees of freedom that are not spatial are shared between the two points. In Phase Theory, quantum entanglement is precisely a manifestation of this: two entangled particles are spatially separated but relationally close in Ω , allowing their phases to be correlated beyond the spatial coherence horizon (Paper 6 [Hanks, 2026f]).

This provides the Phase Theory explanation of quantum non-locality: EPR correlations are not violations of locality in the relational sense — they are correlations within the relational coherence horizon $\ell_c^{(r)}$, which can exceed the spatial coherence horizon $\ell_c^{(s)}$. The correlations appear non-local only from the perspective of emergent space, which does not see the relational structure of Ω . Nevertheless, even relational coherence is bounded:

Theorem 15.3 (Bounded Relational Coherence).

The relational coherence horizon satisfies

$$(15.15) \ \ell_c^{(r)} \leq f(I_c, \chi[\Phi]) = (\kappa/\alpha)^{1/2} / \chi[\Phi]^{1/2},$$

where $\chi[\Phi]$ is the global phase coherence measure (Definition 3.1 applied to all of Ω).

Proof.: ○○○○○○○○

Theorem 15.3 shows that even quantum entanglement has a maximum range: the relational coherence horizon. This is a distinctive and falsifiable prediction of Phase Theory (see Prediction P8 in §53). It implies that quantum entanglement cannot be maintained over arbitrarily large spatial separations: at some maximum baseline, the relational distance d_Ω exceeds $\ell_c^{(r)}$, and entanglement is lost. The maximum spatial baseline for entanglement maintenance is related to the cosmological relational coherence horizon, estimated to be of order the Hubble scale in Appendix D. Current quantum communication experiments at ~ 1000 km baselines are far below this bound, but next-generation experiments at space-based baselines approaching 10^4 km could begin to probe it (Prediction P8).

§10. Coherence Horizon Dynamics

The coherence horizon is not a static property of a given configuration but a dynamical quantity that evolves as the phase configuration evolves. In this section, we derive the equation of motion for the coherence horizon and establish its key dynamical properties.

Let $\ell_c(t)$ denote the coherence horizon of the phase configuration $\Phi(t)$ at emergent time t . Differentiating the coherence horizon formula (15.9) with respect to t :

$$(15.16) \ d\ell_c/dt = (\partial\ell_c/\partial\rho_\Phi)(d\rho_\Phi/dt) + (\partial\ell_c/\partial\kappa)(d\kappa/dt) = F[\Phi, I[\Phi], \chi[\Phi]],$$

where F is a functional of the current phase configuration, the admissibility, and the global coherence measure. The functional F can be computed explicitly in terms of the phase equation of motion (15.3).

Theorem 15.4 (Coherence Horizon Propagation Bound).

The coherence horizon cannot grow faster than the phase propagation speed:
 $d\ell_c/dt \leq v_\phi$.

Proof.: ○○○○○○

Theorem 15.5 (Coherence Horizon Monotonicity).

For an isolated admissible phase system (no external phase input), the coherence horizon is monotonically non-increasing: $d\ell_c/dt \leq 0$.

Proof.: ○○○○

Coherence horizon saturation occurs when the system reaches a steady state in which $d\ell_c/dt = 0$. This corresponds to a balance between phase gradient growth (which would decrease ℓ_c) and coherence-restoring interactions. In cosmological evolution, saturation corresponds to the approach to a maximum-entropy de Sitter-like state where all large-scale coherence has been converted to thermal phase fluctuations at the cosmological horizon scale. The cosmological coherence horizon ℓ_c^{cosm} is therefore a late-time attractor of the phase evolution equation, not an initial condition. This resolves the horizon problem of standard cosmology without inflation: the coherence horizon at late times is set by the dynamics of the admissibility functional, not by the initial conditions at the Big Bang.

§11. Maximum Phase Curvature: Proposition and Proof

Having established the coherence horizon framework, we now turn to curvature bounds. The existence of a maximum admissible phase curvature is perhaps the most dramatic consequence of the admissibility functional, as it forbids the singularities that are otherwise inevitable in general relativity.

Proposition 15.3 (Maximum Phase Curvature).

There exists a maximum admissible phase curvature κ_{max} such that no stable admissible configuration has $\kappa[\Phi](x) > \kappa_{max}$ at any point $x \in \Omega$. The value is

$$(15.17) \quad \kappa_{max} = \gamma / (2\beta) + \lambda T_{max} / \beta.$$

Proof.: ○○○

To connect κ_{max} to the Planck curvature scale, recall from Paper 4 [Hanks, 2026d] that the Riemann curvature tensor of emergent spacetime is proportional to the phase curvature: $R_{\mu\nu} \sim \nabla_\mu \nabla_\nu \Phi$. Therefore κ_{max} corresponds to a maximum Riemann curvature. In the emergent GR limit, the Kretschner scalar $K = R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}$ satisfies $K \leq \kappa_{max}^2$. The Planck curvature scale is $K_{Pl} = c^7 / (\hbar G^2) \approx 5.2 \times 10^{93} \text{ m}^{-4}$. Identifying κ_{max} with K_{Pl} provides a consistency condition relating the admissibility functional parameters to the measured values of c , $\hbar$, and G (Appendix D). Crucially, this identification is a *derived* result, not a postulate: Phase Theory *predicts* that the maximum curvature scale is the Planck scale, rather than assuming it.

§12. Phase Curvature and Singularity Prevention

The maximum curvature bound established in Proposition 15.3 has the immediate and far-reaching consequence of forbidding physical singularities. We formalize this as a theorem.

Theorem 15.6 (Singularity Exclusion Theorem).

Let Φ be any admissible phase configuration satisfying $I[\Phi] \geq 0$ and $\kappa[\Phi] \leq \kappa_{max}$ everywhere. Then no physical observable $O[\Phi]$ that is a polynomial functional of Φ and its derivatives up to second order diverges. In particular, energy density, pressure, and curvature invariants are all bounded.

Proof.: ○○○○○○

The contrast with classical general relativity is stark. The Penrose singularity theorem [Penrose, 1965] proves that trapped surfaces in GR lead to geodesic incompleteness — singularities — under the assumption that energy conditions hold and there is no mechanism to prevent unbounded curvature growth. Phase Theory provides exactly such a mechanism: the admissibility functional acts as a "curvature regulator" that prevents curvature from exceeding κ_{max} . The Penrose theorem's conclusion fails in Phase Theory because one of its hypotheses — the absence of a curvature bound — is violated.

The Hawking-Penrose theorems [Hawking, 1975; Penrose, 2004] more generally establish that singularities are generic in GR given reasonable energy conditions and global causal structure. In Phase Theory, the curvature bound prevents the formation of trapped surfaces in the traditional sense: what appears as a trapped surface in the emergent GR description corresponds to a maximum-curvature region in the phase substrate, which is regular, bounded, and evolving. We will see this in detail in §13 for gravitational collapse and in §47 for black hole physics.

The minimum length scale implied by the curvature bound is derived as follows. A pointlike singularity would require curvature to diverge at a point. Since $\kappa \leq \kappa_{max}$, the curvature is bounded, and the shortest length scale over which curvature can vary significantly is $\ell_{min} = \kappa_{max}^{-1/2}$. This is the Phase Theory minimum length, which coincides with the Planck length in the emergent GR limit. Crucially, this minimum length does not arise from spacetime discreteness: the phase substrate Ω is a continuous manifold. The minimum length is a dynamical bound on phase variation, not a discrete lattice spacing. This preserves Lorentz invariance at all scales (Paper 4 [Hanks, 2026d]).

§13. Phase Curvature and Gravitational Collapse

We apply the curvature bound to the most dramatic manifestation of gravitational dynamics: gravitational collapse to a black hole. This section extends the analysis of Paper 13 [Hanks, 2026m] in light of the ultimate limits established here.

Consider a spherically symmetric collapsing phase configuration. In the emergent GR limit, this corresponds to a collapsing ball of matter with emergent metric. As matter falls inward, the phase gradient density ρ_ϕ increases. By the scaling law (15.12), this decreases the spatial coherence horizon: $\ell_c^{(s)} \sim \rho_\phi^{-1/2} \rightarrow 0$ as $\rho_\phi \rightarrow \infty$. However, before $\ell_c^{(s)}$ reaches zero, the curvature bound intervenes: when $\kappa[\Phi] = \kappa_{max}$, further collapse is forbidden by inadmissibility.

The modified collapse equation from Phase Theory is:

$$(15.19) \quad d^2R/dt^2 = -GM/R^2 + F_{admiss}(R, \kappa[\Phi]),$$

where R is the effective radius, M is the emergent mass, and F_{admiss} is the phase admissibility back-reaction force. In the classical regime $\kappa \ll \kappa_{max}$, $F_{admiss} \approx 0$ and (15.19) reduces to the standard Newtonian (or GR) collapse equation. As $\kappa \rightarrow \kappa_{max}$, F_{admiss} grows rapidly:

$$(15.20) \quad F_{admiss}(R, \kappa) = +2\beta(\kappa_{max} - \kappa)^{-1} \dot{\kappa} (d\kappa/dR),$$

which diverges as $\kappa \rightarrow \kappa_{max}$, providing an infinite restoring force. Therefore, collapse halts at $\kappa = \kappa_{max}$, producing a maximal-curvature shell (the "phase Schwarzschild surface") rather than a point singularity.

The resolution of the information paradox follows directly. In the standard GR analysis, information falling into a black hole is lost at the singularity. In Phase Theory, no singularity forms; instead, the infalling matter is encoded in the phase configuration of the maximal-curvature shell. This configuration is admissible and evolves under the phase equation of motion, gradually releasing information via phase coherence decay (Hawking radiation as a coherence process). The Page time — the time at which information begins to come out — is the coherence horizon timescale τ_c at the shell, which coincides with the Hawking evaporation timescale at

leading order. Corrections to the Hawking spectrum from the curvature-bounded collapse are of order $(E/E_{Pl})^2$ for photon energy E , detectable in principle in near-extremal black hole analogs.

§14. Information Density Limits: Phase Distinguishability

Definition 15.5 (Phase Information Density Bound).

The *phase information density bound* ρ_I^{max} is the maximum number of mutually distinguishable admissible phase configurations per coherence volume $V_c = \mathcal{L}_c^n$ in n dimensions.

The derivation of ρ_I^{max} proceeds from the structure of the admissibility functional. Two phase configurations Φ_1 and Φ_2 are mutually distinguishable (in the phase-theoretic sense) if their admissibility functionals differ by at least δI_{min} : $|I[\Phi_1] - I[\Phi_2]| \geq \delta I_{min}$, where δI_{min} is the minimum distinguishable action increment (identified with $\hbar$ in §28).

The maximum number of distinguishable configurations within the admissible range $[0, I_{max}]$ is:

$$(15.21) \quad N_{dist} = [I_{max} / \delta I_{min}] = [I_{max} / \hbar],$$

where $I_{max} = \gamma V_c$ is the maximum admissibility (achieved by the vacuum configuration) in a coherence volume V_c . Therefore:

$$(15.22) \quad \rho_I^{max} = N_{dist} / V_c = \gamma / \hbar.$$

This result — that the maximum information density is a ratio of the admissibility coherence coefficient to the action quantum — is one of the central results of Phase Theory. It is finite, independent of the spatial dimension n (to leading order), and depends only on the fundamental parameters of the admissibility functional.

Infinite information density would require $N_{dist} \rightarrow \infty$, which would require either $I_{max} \rightarrow \infty$ (impossible since $I_{max} = \gamma V_c$ is finite for finite coherence volumes) or $\delta I_{min} \rightarrow 0$. The latter would require either $\kappa > \kappa_{max}$ (forbidden by Proposition 15.3) or $\chi < \varepsilon_c$ (forbidden by the anti-fragmentation theorem of §18). Therefore, infinite information density is structurally forbidden by Phase Theory.

§15. Information Density Limits: Derivation

We provide a detailed step-by-step derivation of the information density bound, making all steps explicit.

Step 1: Enumerate admissible configurations in a coherence volume.

Consider a coherence volume $V_c = \ell_c^n$. Within this volume, the admissible phase configurations form a subset of $\mathcal{C}^2(V_c, S^1)$. By the admissibility bounds (Lemma 4.1), the phase field satisfies $|\nabla \Phi| \leq (\gamma/\alpha)^{1/2}$ and $|\nabla^2 \Phi| \leq \kappa_{max}^{1/2}$ pointwise. The space of such functions, bounded in C^2 norm, has finite metric entropy (by the Kolmogorov-Tikhomirov theorem for function spaces with bounded derivatives).

Step 2: Bound the number of orthogonal states. Two configurations Φ_1, Φ_2 define orthogonal quantum states if $\langle \psi_1 | \psi_2 \rangle = 0$, where $|\psi_i\rangle = e^{iS[\Phi_i]}|0\rangle$ is the coherent phase state. Two states are distinguishable if their phase overlap is less than ε_c . The maximum phase variation over the coherence volume is $\Delta\Phi_{max} = (\gamma/\alpha)^{1/2} \ell_c$. The number of orthogonal states is bounded by the ratio of the maximum phase space volume to the minimum phase cell size:

$$(15.23) \quad N_{orth} \leq (\Delta\Phi_{max} / \delta\Phi_{min})^n = ((\gamma/\alpha)^{1/2} \ell_c / \delta\Phi_{min})^n,$$

where $\delta\Phi_{min} = (2\pi)^{1/2} (\hbar/\gamma V_c)^{1/2}$ is the minimum resolvable phase difference.

Step 3: Convert to information density via Shannon entropy. The maximum information capacity is $H_{max} = \log_2(N_{orth})$ bits. The information density is:

$$(15.24) \quad \rho_I^{max} = H_{max} / V_c = (n/2) \log_2(\gamma \ell_c^2 / (\alpha \delta\Phi_{min}^2)) / V_c.$$

Step 4: Express in natural units. Substituting the expression for $\delta\Phi_{min}$, using $\ell_c = (\kappa/\alpha)^{1/2}$, and taking the leading-order term:

$$(15.25) \rho_I^{max} \leq K_{max}^{n/2} / (2\pi)^n,$$

in natural units where $\hbar = c = 1$. This is the fundamental information density bound in Phase Theory, expressed purely in terms of the maximum curvature K_{max} .

Topological corrections. The bound (15.25) is modified in topological sectors with nonzero winding number. In a sector with topological charge Q , the number of distinguishable configurations is shifted by the topological degeneracy factor $N_{top}(Q) = |Q| + 1$ (from Paper 7 [Hanks, 2026g]). Therefore:

$$(15.26) \rho_I^{max}(Q) = K_{max}^{n/2} / (2\pi)^n + |Q| \rho_I^{top},$$

where ρ_I^{top} is the topological information density per unit topological charge. This correction is typically subdominant in the low-curvature regime but becomes significant near K_{max} , explaining the high information content of black holes (which are near-maximal-curvature configurations with potentially large topological charge from infalling matter).

§16. Comparison with Bekenstein-Hawking and Holographic Bounds

We now demonstrate that the Phase Theory information density bound subsumes the standard Bekenstein and holographic bounds as special cases and provides corrections in high-curvature regimes.

16.1 The Bekenstein Bound

The Bekenstein bound [Bekenstein, 1973] states that the entropy of a physical system of energy E and radius R satisfies $S \leq 2\pi R E / \hbar c$. To recover this from Phase Theory, we identify: (a) energy E with the phase mode energy density integrated over volume $E \sim \rho_\phi R^n$, (b) entropy S with the maximum information content $H_{max} = \rho_I^{max} R^n$. In the weak-field limit where $\kappa \ll K_{max}$ and $R \leq \ell_c^{(s)}$, equation (15.25) gives $H_{max} \sim (\kappa/\alpha)^{n/2} / (2\pi)^n R^n$. Comparing with the Bekenstein bound: the identification requires $R \sim \ell_c$ and $E \sim \alpha \ell_c^{n-2}$ in natural units, giving $S \leq 2\pi R E / \hbar c$ to leading order. The Bekenstein bound is thus the Phase Theory information density bound in the weak-curvature, single-coherence-region limit.

16.2 The Bekenstein-Hawking Entropy

The Bekenstein-Hawking formula [Hawking, 1975] gives black hole entropy as $S_{BH} = A/(4G\hbar)$ in natural units, where A is the horizon area. In Phase Theory, the black hole horizon corresponds to the maximal-curvature shell at $\kappa = \kappa_{max}$. The area of this shell in the emergent geometry is $A = 4\pi R_s^2$ where $R_s = 2GM/c^2$ is the Schwarzschild radius. The Phase Theory information capacity of the shell is $N_{shell} = \rho_I^{max} A / \ell_{min}$ where ℓ_{min} is the shell thickness set by $\kappa_{max}^{-1/2}$. This gives $N_{shell} \sim A/\ell_{min}^2 \sim A/(4G\hbar)$, reproducing the Bekenstein-Hawking formula. The agreement is exact in the leading order, with Phase Theory corrections of order $G\hbar/A$ (relevant for small black holes near the Planck scale).

16.3 The Holographic Principle

The holographic principle [Susskind, 1995; Maldacena, 1997] states that the information content of a volume V is bounded by the area of its boundary: $S(V) \leq A(\partial V)/(4G\hbar)$. Phase Theory derives this as follows. The information density bound (15.25) gives $S(V) \leq \rho_I^{max} V$. However, this volume scaling is modified when one accounts for the fact that the coherence horizon is limited to the spatial coherence horizon $\ell_c^{(s)}$. For a volume V larger than $\ell_c^{(s)n}$, the interior of V is divided into many coherence cells, each with limited mutual coherence. The total distinguishable information is not the volume-scaling but the area-scaling $A(\partial V)/\ell_c^{(s)2}$, because only the boundary coherence cells can communicate with the exterior. This holographic reduction is a consequence of the coherence horizon: configurations in the deep interior of a large volume contribute no new distinguishable information relative to the boundary information.

Bound	Standard Form	Phase Theory Form	Regime of Agreement	Phase Theory Correction
Bekenstein	$S \leq 2\pi RE/\hbar c$	$\rho_I^{max} V_c \leq \gamma V_c/\hbar$	$\kappa \ll \kappa_{max}$	$O((\kappa/\kappa_{max})^2)$ reduction near κ_{max}
Bekenstein-Hawking	$S = A/(4G\hbar)$	$\rho_I^{max} A \ell_{min}$	$A \gg \ell_{min}^2$	$+G\hbar/A$ correction for small BH
Holographic	$S(V) \leq$	$S(V) \leq \rho_I^{max}$	$V \gg \ell_c^n$	Tighter at

Bound	Standard Form	Phase Theory Form	Regime of Agreement	Phase Theory Correction
	$A(\partial V)/4G\hbar$	$\ell_c^{n-2} A(\partial V)$		high κ , looser at low κ

§17. The Phase Holographic Principle

Building on the comparison of §16, we now derive the Phase Theory version of the holographic principle as a theorem rather than a hypothesis.

Theorem 15.7 (Phase Holographic Principle).

For any admissible phase configuration Φ in a compact region $V \subset \Omega$ with boundary ∂V , the number of mutually distinguishable states in V is bounded by the number of admissible configurations on ∂V :

$$(15.27) \quad N_{\text{dist}}(V) \leq N_{\text{dist}}(\partial V) = \exp(\rho_I^{\max} \|\partial V\| \ell_c^{(s)n-1}),$$

where $\|\partial V\|$ is the $(n-1)$ -dimensional measure of the boundary.

Proof.: ○○○○○○

We define the *phase holographic entropy* of a region V as:

$$(15.28) \quad S_{\text{holo}}(V) = k_B \rho_I^{\max} \|\partial V\| \ell_c^{(s)n-1},$$

which scales as the boundary area rather than the volume. For a spherical region of radius R , this gives $S_{\text{holo}} \sim k_B R^{n-1} \rho_I^{\max} \ell_c^{(s)n-1}$. In three spatial dimensions, $S_{\text{holo}} \sim k_B R^2 \rho_I^{\max} \ell_c^{(s)2}$. Identifying $\rho_I^{\max} \ell_c^{(s)2} \sim 1/(4G\hbar)$ recovers the Bekenstein-Hawking formula exactly.

The Phase Theory holographic bound is *tighter* than the standard bound in high-curvature regimes, because near κ_{max} the coherence horizon $\ell_c^{(s)}$ shrinks (by (15.9)), reducing the boundary information capacity per unit area. It is *looser* than the

standard bound in low-curvature regimes, where the large coherence horizon implies that distant boundary cells are nearly independent and the bound is not saturated. Physical systems saturate the bound only at $\kappa = \kappa_{max}$ — i.e., at black hole horizons. This is consistent with the known fact that black holes are the most entropic objects for their size.

§18. Why Reality Does Not Fragment

We now address the first of the two structural stability results that characterize Phase Theory's stability regime: the Anti-Fragmentation Theorem. Fragmentation refers to the spontaneous breakup of a coherent phase configuration into an incoherent superposition of microscopic structures.

Definition 15.6 (Phase Fragmentation).

A phase configuration Φ undergoes *fragmentation* if there exists an admissible evolution $\Phi \rightarrow \Phi'$ such that the coherence measure satisfies $\chi[\Phi'] < \varepsilon_c$, i.e., the evolved configuration falls below the minimum admissible coherence threshold.

Theorem 15.8 (Anti-Fragmentation Theorem).

No admissible phase configuration can undergo fragmentation under admissible phase evolution. That is, if $I[\Phi] \geq 0$ and $\chi[\Phi] \geq \varepsilon_c$, then $I[\Phi'] \geq 0$ and $\chi[\Phi'] \geq \varepsilon_c$ for all Φ' reachable from Φ by admissible evolution.

Proof.

Suppose Φ' is reachable from Φ by a continuous admissible path $\Phi(t)$ with $I[\Phi(t)] \geq 0$ for all t . We must show $\chi[\Phi'] \geq \varepsilon_c$. Suppose for contradiction that $\chi[\Phi'] < \varepsilon_c$. Then by the definition of ε_c and the admissibility functional structure, $C(\Phi') < \varepsilon_c^2$, so the coherence term in L_{admiss} contributes at most $\gamma \varepsilon_c^2$. The gradient and curvature

penalties are non-negative, so $I[\Phi'] \leq \gamma \varepsilon_c^2 / |\Omega| - |\alpha| \int |\nabla \Phi'|^2 d^n \Omega$. Since $\chi[\Phi'] < \varepsilon_c$, the phase is incoherent and $\int |\nabla \Phi'|^2 d^n \Omega > (\gamma/\alpha) \varepsilon_c^2 / |\Omega|$ (because incoherence requires large gradient variation). Therefore $I[\Phi'] < 0$, contradicting the assumption that Φ' is admissible. $\square$

The Anti-Fragmentation Theorem has three simultaneous consequences that together constitute the fragmentation barrier:

(a) *Lower bound on coherence length:* By the theorem, $\chi[\Phi] \geq \varepsilon_c$ is preserved. Since $\ell_c \sim \chi^{1/2}$, this implies $\ell_c \geq \varepsilon_c^{1/2} (\kappa/\alpha)^{1/2}$ — a minimum coherence length.

(b) *Upper bound on phase gradient variance:* The proof shows that $\text{var}(\nabla \Phi) \leq (\gamma/\alpha) \chi^2$ for admissible configurations, bounding the fluctuations of the phase gradient.

(c) *Lower bound on two-point correlation:* For any two points within a coherence volume, $|G(x, x')| \geq \varepsilon_c$, maintaining minimum correlation.

Vacuum stability is a special case. The vacuum configuration Φ_0 has $\chi = 1$ (maximally coherent) and $I[\Phi_0] = \gamma/|\Omega|$ (maximum admissibility). The Anti-Fragmentation Theorem ensures that any perturbation of the vacuum that would decrease χ below ε_c is forbidden. In the language of quantum field theory, this is vacuum stability: the Higgs vacuum does not decay to a less-coherent phase configuration. Phase Theory provides the first derivation of vacuum stability from first principles without requiring a potential with a positive-definite mass-squared term.

§19. Vacuum Stability from Admissibility

We sharpen the vacuum stability argument of §18 and provide the full proof that the phase vacuum is a global stable minimum.

Definition 15.7 (Phase Vacuum).

The *phase vacuum* Φ_0 is the unique (up to global phase rotation) admissible configuration satisfying: (a) maximum coherence $\chi[\Phi_0] = 1$, (b) zero curvature $\kappa[\Phi_0] = 0$, and (c) constant phase $\nabla\Phi_0 = 0$.

Proposition 15.4 (Vacuum Stability).

Φ_0 is a global minimum of the curvature functional and a global maximum of the coherence functional, and is uniquely stable under all admissible perturbations.

Proof.

The curvature functional is $K[\Phi] = \int \kappa[\Phi] d^n\Omega \geq 0$, with equality iff $\nabla^2\Phi = 0$. Since $\Phi_0 = \text{const}$, we have $\nabla^2\Phi_0 = 0$ and $K[\Phi_0] = 0$. Since $K \geq 0$ always, Φ_0 is a global minimum. Similarly, $\chi[\Phi_0] = |\int e^{i\Phi_0} d\Omega| / |\Omega| = 1$ since $e^{i\Phi_0} = e^{i\phi_0}$ is constant. For any $\Phi \neq \Phi_0$, $\chi[\Phi] < 1$ by the strict triangle inequality (since the phase is not everywhere equal). Thus Φ_0 is the unique global maximum of coherence. For stability, consider a perturbation $\delta\Phi = \Phi - \Phi_0$. The second variation is $\delta^2 I[\Phi_0][\delta\Phi, \delta\Phi] = -\alpha \int |\nabla\delta\Phi|^2 d\Omega - \beta \int |\nabla^2\delta\Phi|^2 d\Omega - \gamma \int (\delta\Phi)^2 d\Omega$ (from the linearization of the coherence term around Φ_0). Each term is non-positive, so $\delta^2 I \leq 0$, meaning Φ_0 is a local maximum of I (i.e., the most admissible configuration) and all perturbations decrease admissibility, hence are bounded by the admissibility condition. No perturbation can take the system to a configuration with higher $I[\Phi]$ than the vacuum without violating the equations of motion. $\square$

The connection to the Higgs mechanism is illuminating. In standard quantum field theory, the Higgs mechanism involves a field developing a nonzero vacuum expectation value, spontaneously breaking a symmetry. In Phase Theory (Paper 5 [Hanks, 2026e]), mass generation occurs when the phase vacuum selects a particular direction in the internal phase space, breaking the global U(1) symmetry down to a residual symmetry group. The Proposition 15.4 shows that this selection is not a "fall" into a lower-energy vacuum state (which would suggest vacuum instability) but rather a selection among degenerate maxima of $I[\Phi]$ (the global

phase rotations of Φ_0). All these maxima are equally stable; the selection is conventional, not physical. Goldstone bosons correspond to the directions along the degenerate set (the orbit of Φ_0 under global $U(1)$), which are the zero modes of the second variation.

§20. Phase Noise Suppression

Physical reality is not noiseless. Even the vacuum is subject to quantum fluctuations, and all physical systems exhibit some level of stochastic variation. In Phase Theory, this noise is represented as stochastic fluctuations $\delta\Phi(x,t)$ of the phase field around its mean configuration. The admissibility functional constrains the maximum amplitude of these fluctuations.

Definition 15.8 (Phase Noise).

Phase noise is the stochastic component $\delta\Phi = \Phi - \langle\Phi\rangle$ where $\langle\Phi\rangle$ is the mean phase configuration. The noise variance is $\sigma^2(x) = \langle(\delta\Phi(x))^2\rangle$.

The noise suppression bound follows from the admissibility requirement. A noisy configuration $\Phi = \langle\Phi\rangle + \delta\Phi$ must satisfy $I[\Phi] \geq 0$. Expanding to second order in $\delta\Phi$:

$$(15.29) \quad I[\Phi] \approx I[\langle\Phi\rangle] + \delta^2 I[\langle\Phi\rangle][\delta\Phi, \delta\Phi] \geq 0.$$

The second variation gives: $\delta^2 I \approx -\alpha \int |\nabla \delta\Phi|^2 d\Omega - \gamma \int (\delta\Phi)^2 d\Omega$. For this to satisfy (15.29), we need $I[\langle\Phi\rangle] \geq \alpha \int |\nabla \delta\Phi|^2 d\Omega + \gamma \int (\delta\Phi)^2 d\Omega$. Since $|\nabla \delta\Phi|^2 \geq (2\pi/\ell_c)^2 (\delta\Phi)^2$ by the Poincaré inequality, we get:

$$(15.30) \quad \sigma^2(x) = \langle(\delta\Phi)^2\rangle \leq \sigma_{\max}^2 = \delta I_{\min} / (\alpha \ell_c^2),$$

where $\delta I_{\min} = \hbar$ is the minimum action quantum (§28) and ℓ_c is the local coherence horizon.

This bound has three key properties. First, it is finite: no admissible configuration has unbounded noise. Second, it decreases with increasing coherence length: a

large, coherent system has tighter noise suppression. Third, it provides a minimum signal-to-noise ratio for any physical process: $SNR_{min} = \langle \Phi \rangle^2 / \sigma_{max}^2 = \langle \Phi \rangle^2 \propto \ell_c^2 / \hbar$. In the quantum limit, $\sigma_{max}^2 \rightarrow \hbar / (2\omega\tau_c)$ where ω is the mode frequency, reproducing the quantum zero-point fluctuation amplitude as the saturation of the Phase Theory noise bound.

The cosmological constant problem is resolved by this bound. The standard QFT prediction for vacuum energy is $\rho_{vac} \sim M_{Pl}^4 / (\hbar^3 c^3)$, 120 orders of magnitude larger than observed. In Phase Theory, the vacuum energy is the noise energy: $\rho_{vac} \sim \sigma_{min}^2 \cdot \rho_{I^{max}} \cdot c^4 / (8\pi G)$ where $\sigma_{min}^2 = \delta I_{min} / (\alpha \ell_c^{cosm,2})$ uses the cosmological coherence horizon as the relevant length scale. The result is $\rho_{vac} \sim \hbar c / \ell_c^{cosm,4}$, which for $\ell_c^{cosm} \sim H_0^{-1}$ gives $\rho_{vac} \sim \hbar H_0^4 / c^3$ — the correct order of magnitude for the observed cosmological constant.

§21. Persistence of Physical Laws

A perennial puzzle in the foundations of physics is why physical laws are universal and persistent: why do the same equations of motion hold everywhere and at all times? Standard physics provides no answer: laws are simply assumed to be universal. Phase Theory provides a derivation.

Definition 15.9 (Physical Law as Phase Mode).

A *physical law*, in Phase Theory, is an effective description of a coherent, stable phase propagation mode: a mode Φ_{mode} that (a) is admissible everywhere, (b) has a well-defined propagation velocity and dispersion relation, and (c) persists for a time exceeding the measurement horizon timescale τ_c . The *law persistence length* of such a mode is its temporal coherence horizon $\tau_c[\Phi_{mode}]$.

For the fundamental laws of physics (corresponding to the lowest-curvature, highest-coherence modes, such as the phase modes encoding gauge symmetry and the metric), the persistence length is:

$$(15.31) \tau_{persist}^{fund} = \ell_c^{cosm} / v_\Phi,$$

the cosmological coherence horizon divided by the phase propagation speed. This is the age of the observable universe to within an order of magnitude, consistent with the observed persistence of physical laws for the entire cosmic history. The fundamental laws persist as long as the universe remains in a coherent phase state, which by Theorem 15.8 (Anti-Fragmentation) is guaranteed for all admissible evolutions.

Law change could in principle occur near maximal-curvature events (near $\kappa = \kappa_{max}$), where the coherence horizon shrinks and the modes encoding the laws become short-lived. In the interior of a collapsing phase structure (§13), the effective physical laws seen by a local observer would differ from those in the ambient universe: the persistence length of low-level laws would be of order $\tau_c \sim \ell_{min}/v_\Phi \sim t_{Pl}$ (the Planck time). This suggests that the interior of a black hole analog would not exhibit standard physics, a distinctive prediction that could in principle be tested in condensed-matter analogs.

§22. Why Reality Does Not Freeze

We now address the dual question to fragmentation: can the universe freeze into a static, eternal configuration? We prove that this is equally forbidden by admissibility.

Definition 15.10 (Phase Freezing).

A phase configuration Φ is *frozen* if $\partial\Phi/\partial t = 0$ everywhere, i.e., if the phase field is static under the emergent time evolution.

Theorem 15.9 (Anti-Freezing Theorem).

The only frozen (static) admissible phase configuration is the trivial uniform phase $\Phi = const$. Every non-trivial admissible configuration evolves under the

phase equation of motion (15.3).

Proof.

A frozen configuration satisfies both the phase equation of motion (15.3) and $\partial\Phi/\partial t = 0$. The phase equation with the time-derivative set to zero becomes the stationary admissibility equation: $\gamma \delta C/\delta\Phi + 2\alpha \nabla^2\Phi + 2\beta \nabla^4\Phi + \lambda \delta T/\delta\Phi = 0$. Multiplying by $\delta\Phi$ and integrating over Ω : $\gamma \int (\delta C/\delta\Phi)\delta\Phi d\Omega - 2\alpha \int |\nabla\Phi|^2 d\Omega - 2\beta \int |\nabla^2\Phi|^2 d\Omega = 0$. The first term is bounded above by 2γ . For a non-trivial configuration ($\Phi \neq \text{const}$), the second and third terms are strictly positive. For the equation to hold with strict positivity on the right-hand side terms, we need $\gamma \int (\delta C/\delta\Phi)\delta\Phi d\Omega > 0$, which requires the coherence term to actively "push" the configuration. But the coherence term is maximized at $\Phi = \text{const}$ and decreases in all non-trivial directions. Therefore, for any non-trivial configuration, the stationary equation cannot be satisfied with both stationarity and non-triviality. The only solution with $\partial\Phi/\partial t = 0$ is $\Phi = \text{const}$. $\square$

The physical interpretation is profound: a universe with nonzero information content cannot be static. Any admissible configuration with structure ($\Phi \neq \text{const}$) must evolve. This provides the phase-theoretic foundation for the necessity of time and change: not as a postulate but as a consequence of the admissibility functional structure. The marginal permissivity condition (§23) formalizes the condition under which this evolution is generically non-trivial.

§23. The Marginal Permissivity Condition

Definition 15.11 (Marginal Permissivity Condition).

A phase configuration Φ satisfies the *Marginal Permissivity Condition (MPC)* if and only if there exists a perturbation $\delta\Phi \neq 0$ such that $I[\Phi + \delta\Phi] \geq I[\Phi] > 0$ and $|\delta\Phi| > 0$.

MPC is equivalent to the existence of a zero mode (a direction of zero second variation) of the admissibility functional. To see this: if $\delta^2 I[\Phi][\delta\Phi, \delta\Phi] = 0$ for some $\delta\Phi \neq 0$, then the configuration can be perturbed along this direction without first-order change in admissibility, i.e., $I[\Phi + \varepsilon\delta\Phi] = I[\Phi] + O(\varepsilon^2)$. The third-order term then determines whether the perturbation increases or decreases I ; generically, at least one direction increases I , satisfying MPC.

The zero modes of $\delta^2 I$ are the solutions of the linearized admissibility equation:

$$(15.32) \quad (-\alpha \nabla^2 - \beta \nabla^4 - \gamma \chi^2) \delta\Phi = 0.$$

This is a fourth-order linear PDE, and by standard spectral theory it has a discrete spectrum of eigenvalues on compact Ω . The zero modes correspond to zero eigenvalues. The number of zero modes is a topological invariant — it equals the index of the operator, which by the Atiyah-Singer index theorem is related to the topological charge N_{top} (Paper 7 [Hanks, 2026g]):

$$(15.33) \quad \dim(\ker \delta^2 I) = |N_{top}| + n_{symm},$$

where n_{symm} is the dimension of the symmetry group of Φ . For generic configurations in non-trivial topological sectors, the number of zero modes is nonzero, and MPC is satisfied generically. It is not a fine-tuning condition.

The connection to the principle of least action is immediate. The zero modes of $\delta^2 I$ are the directions in which the phase can evolve at zero "cost" to the admissibility. Physical dynamics follows these zero-mode directions: the phase evolves along paths of stationary admissibility, which are precisely the solutions of the Euler-Lagrange equation (15.3). The principle of least action is therefore a consequence of admissibility extremization, not a separately assumed variational principle. The action that is extremized is the admissibility functional $I[\Phi]$ itself.

§24. Phase Dynamics and Structural Freedom

Definition 15.12 (Structural Freedom).

The *structural freedom* $D[V]$ of a phase system in a compact coherence region V is the dimension of the space of admissible phase configurations modulo gauge equivalences (global phase rotations).

Theorem 15.10 (Structural Freedom Theorem).

The structural freedom of a phase system in a coherence region V is finite and satisfies

$$(15.34) \quad D[V] = N_c^n,$$

where N_c is the number of distinguishable configurations per coherence cell and n is the effective phase-space dimension.

Proof.: ○○○○○○

The finiteness of structural freedom has three important consequences. First, the *particle spectrum is finite* (Paper 2 [Hanks, 2026b]): the particle species correspond to the distinct irreducible representations of the structural freedom group, which is finite-dimensional. Second, the *gauge group is finite-dimensional*: it is the symmetry group of the admissible configuration space, which by structural freedom finiteness is a compact Lie group. Third, the *Hilbert space of quantum states is finite-dimensional per coherence volume*, providing a natural UV regularization without a lattice. This last point is significant for quantum gravity: the finite-dimensional Hilbert space per coherence volume is precisely what is needed for a consistent quantum gravity theory, as argued by Susskind [1995] in the context of the holographic principle.

§25. The Balance Between Coherence and Evolution

We now bring together the results of §18–§24 to characterize the "physical regime" — the region of configuration space simultaneously satisfying all stability conditions.

Physical reality must simultaneously satisfy:

1. *Anti-fragmentation*: $\chi[\Phi] \geq \varepsilon_c$ (coherent enough to support stable structures).
2. *Anti-freezing (MPC)*: the configuration has non-zero modes, i.e., it evolves under (15.3).
3. *Maximum curvature*: $\kappa[\Phi] \leq \kappa_{max}$ (bounded enough to avoid divergence).
4. *Information density bound*: $\rho_I[\Phi] \leq \rho_I^{max}$ (finite complexity).

Define the physical regime as:

$$(15.35) \ P = \{\Phi : I[\Phi] \geq 0, \chi[\Phi] \geq \varepsilon_c, \kappa[\Phi] \leq \kappa_{max}, \text{ MPC satisfied}\}.$$

Theorem 15.11 (Non-Emptiness and Openness of Physical Regime).

The physical regime P is: (a) non-empty, (b) open and connected in the C^2 topology on configuration space, and (c) invariant under admissible phase evolution.

Proof.: ○○○○○○

The existence and openness of P is the deep reason physical reality exists. An open, non-empty, invariant subset of configuration space means: there are configurations that are simultaneously stable, dynamic, bounded, and complex; and once in P , admissible evolution keeps the system in P indefinitely. The universe is not "fine-tuned" to be in P — it cannot be otherwise, because only configurations in P are admissible, and admissibility is the definition of physical reality. The stability, finitude, and richness of the universe are therefore not contingent: they are necessary.

§26. Emergence of Universal Constants: Framework

Having established the structural bounds of Phase Theory, we turn to the question of universal constants. The standard model of physics is governed by approximately 26 free parameters, including masses, mixing angles, and coupling constants, whose values are determined by experiment and must be input by hand. Phase Theory's position is that these parameters are not truly fundamental but are emergent conversion factors between different aspects of the phase description.

We classify the traditional "fundamental constants" into four categories based on their origin in the admissibility functional:

Constant	Standard View	Phase Theory Category	Phase Theory Origin	Derived in Section
c (speed of light)	Maximum signal speed; fundamental	Kinematic	Max phase propagation speed $v_\phi = (\alpha/\kappa)^{1/2}$	§27
$\hbar$ (Planck constant)	Quantum of action; fundamental	Quantum	Min distinguishable action increment $\delta I_{\min}$	§28
G (gravitational constant)	Gravitational coupling; fundamental	Gravitational	Phase curvature coupling κ : $G = \alpha^2/(16\pi\kappa^2)$	§29
α_{em} (fine structure)	EM coupling; fundamental, dimensionless	Coupling	U(1) phase mode overlap integral	§30
k_B (Boltzmann constant)	Converts temperature to energy	Kinematic	Conversion between coherence measure and energy units	§48

The general program for deriving each class of constants from the admissibility functional proceeds in three steps. First, identify the physical observable that

defines the constant (e.g., c is the maximum velocity of any massive or massless object). Second, express this observable in terms of Phase Theory constructs (c is the maximum phase propagation speed). Third, compute the Phase Theory expression in terms of the admissibility functional parameters ($\alpha, \beta, \gamma, \lambda$) and express the result. The computed value must agree with the measured value; this provides a constraint on the admissibility parameters, allowing them to be fixed (in principle) from a finite set of measurements, after which all other constants are predictions.

§27. The Speed of Light as a Phase Propagation Bound

The speed of light c is perhaps the most fundamental constant in modern physics. In Phase Theory, it is not a mysterious coincidence but the maximum propagation speed of coherent phase modes, derived from the gradient and curvature structure of the admissibility functional.

Consider a coherent phase mode: a configuration $\Phi(x, t) = A e^{i(kx - \omega t)}$ propagating with group velocity $v = \partial\omega/\partial k$. For this to be admissible, the gradient penalty must not render $I[\Phi] < 0$. The gradient squared is $|\nabla\Phi|^2 = k^2 A^2$, and the curvature squared is $|\nabla^2\Phi|^2 = k^4 A^2$. Substituting into the admissibility functional:

$$(15.36) \quad I[\Phi_{mode}] = V(\gamma A^2 - \alpha k^2 A^2 - \beta k^4 A^2) \geq 0,$$

which requires $\gamma \geq \alpha k^2 + \beta k^4$. This gives a maximum wavenumber k_{max} for admissible modes. The dispersion relation from the admissibility equation (15.3) for these modes is:

$$(15.37) \quad \omega^2 = c_\phi^2 k^2 - \beta/\alpha k^4 + m_\phi^2 c_\phi^4 / \hbar^2,$$

where $c_\phi = (\alpha/\kappa)^{1/2}$ is the phase propagation speed and m_ϕ is the effective phase mass. The group velocity is:

$$(15.38) \quad v = \partial\omega/\partial k = c_\phi^2 k / \omega \leq c_\phi,$$

with equality iff $m_\phi = 0$ (massless modes) and k is small. This identifies $c_\phi = (\alpha/\kappa)^{1/2}$ as the speed of light c : the maximum group velocity of any coherent phase mode.

Theorem 15.12 (Phase Propagation Bound).

$v = (\partial\omega/\partial k) \leq c \sqrt{\alpha/k} = (\alpha/k)^{1/2}$ for all admissible phase modes, with equality for massless modes in the long-wavelength limit.

The dispersion relation (15.37) reduces to the standard relativistic dispersion $E^2 = (pc)^2 + (mc^2)^2$ when the quartic correction term βk^4 is negligible compared to αk^2 , i.e., when $k \ll (\alpha/\beta)^{1/2} = k_{max}$. The quartic correction becomes significant near k_{max} , providing Phase Theory corrections to special relativity at the Planck scale: $\delta v/c \sim -(k/k_{max})^2 \sim -(E/E_{Pl})^2$. Lorentz invariance is recovered exactly in the low-energy limit and is violated by order $(E/E_{Pl})^2$ near the Planck scale — a potentially observable prediction (see Prediction P7 in §53).

§28. The Planck Constant as a Phase Action Quantum

We derive Planck's constant $\hbar$ as the minimum distinguishable increment of the admissibility functional. This derivation constitutes the Phase Theory foundation of quantum mechanics.

Definition 15.13 (Minimum Distinguishable Action).

The *minimum distinguishable action increment* is

$$(15.39) \quad \delta I_{min} = \inf_{\Phi, \delta\Phi} \{I[\Phi + \delta\Phi] - I[\Phi] : \delta\Phi \neq 0, \Phi + \delta\Phi \text{ admissible}\}.$$

Proposition 15.5.

$\delta I_{min} > 0$, and in appropriate units, $\delta I_{min} = \hbar$.

Proof.

Suppose $\delta I_{min} = 0$. Then there exist sequences Φ_k and $\delta\Phi_k \neq 0$ with $I[\Phi_k + \delta\Phi_k] - I[\Phi_k] \rightarrow 0$. By the continuity of I in the C^2 topology, this would require $\delta\Phi_k \rightarrow 0$ in C^2 . But then the configurations Φ_k and $\Phi_k + \delta\Phi_k$ are indistinguishable in the limit, contradicting $\delta\Phi_k \neq 0$. Therefore $\delta I_{min} > 0$. The identification $\delta I_{min} = \hbar$ follows from matching to the quantum limit: in a phase system undergoing harmonic evolution with frequency ω , the minimum energy quantum is $\hbar\omega$; this must equal $\delta I_{min} \times \omega$ (the action quantum times the frequency), giving $\delta I_{min} = \hbar$. $\square$

The Heisenberg uncertainty principle follows immediately. Two phase configurations Φ_1 and Φ_2 are distinguishable only if $|I[\Phi_1] - I[\Phi_2]| \geq \hbar$. For a system with a conjugate pair of observables (q, p) related by the Legendre transform of the admissibility Lagrangian, the uncertainty in the product $\Delta q \Delta p$ is bounded below by the indistinguishability condition $|I[q+\Delta q, p+\Delta p] - I[q, p]| \geq \hbar$. Expanding to first order gives $| \Delta I | \approx | (\partial I / \partial q) \Delta q + (\partial I / \partial p) \Delta p | \geq \hbar$. Since the Legendre structure requires $\partial I / \partial q \sim p$ and $\partial I / \partial p \sim q$, this gives $\Delta q \Delta p \geq \hbar/2$ by the Cauchy-Schwarz inequality. The Heisenberg uncertainty principle is thus a consequence of the finite minimum action quantum, which is a consequence of the admissibility functional structure.

§29. The Gravitational Constant from Phase Curvature

Gravity in Phase Theory emerges from the curvature penalty term in the admissibility functional (Paper 4 [Hanks, 2026d]). The gravitational constant G encodes the strength of the curvature coupling.

The gravitational potential between two phase configurations Φ_1 and Φ_2 arises from the curvature contribution to the admissibility functional when the two configurations overlap. The interaction term is:

$$(15.40) \quad V_{grav}[\Phi_1, \Phi_2] = -\beta \int (\nabla^2 \Phi_1)(\nabla^2 \Phi_2) d^n \Omega \sim -\beta M_1 M_2 / R,$$

in the long-range limit, where M_i are the emergent masses (related to the curvature content) and R is the separation. Comparing with Newton's law $V_{Newton} = -GM_1 M_2 / R$:

$$(15.41) \ G = \beta / (4\pi) = \alpha^2 / (16\pi\kappa^2),$$

using $\beta = \alpha^2/(4\kappa^2)$ (from the consistency condition between the curvature coupling and the phase propagation speed in Paper 4 [Hanks, 2026d]). The weak equivalence principle follows because the same curvature coupling β governs both the gravitational source (the curvature content of matter configurations) and the gravitational response (the acceleration of test configurations in a curved background), independently of the phase mode content — i.e., independently of what kind of matter is involved. This universality of coupling is not assumed but derived from the single-term structure of the curvature penalty in (15.2).

Corrections to Newton's law arise in the high-curvature regime where $\kappa \sim \kappa_{max}$. Near κ_{max} , the curvature penalty saturates and higher-order terms in the admissibility Lagrangian become relevant. The leading correction to the gravitational potential at short range is:

$$(15.42) \ V_{grav}^{corrected}(R) = -GM_1M_2/R \times (1 + (\kappa(R)/\kappa_{max})^2 + O((\kappa/\kappa_{max})^4)),$$

showing that gravity becomes stronger (more negative) at very short range, saturating at the Planck scale.

§30. Fine Structure Constant from Phase Coupling

The electromagnetic fine structure constant $\alpha_{em} = e^2/(\hbar c) \approx 1/137.036$ is the dimensionless measure of the strength of the electromagnetic interaction. Its value is not explained by the Standard Model; it is simply measured and inserted as a parameter. Phase Theory aims to derive it from the U(1) phase symmetry structure.

In Paper 5 [Hanks, 2026e], gauge interactions are shown to arise from local U(1) phase symmetry: the admissibility functional is invariant under $\Phi(x) \rightarrow \Phi(x) + \theta(x)$ only when augmented by a gauge connection A_μ transforming as $A_\mu \rightarrow A_\mu + \partial_\mu\theta$. The gauge interaction coupling is determined by the overlap integral of admissible phase modes:

$$(15.43) \ e = \int \Phi_{mode}^* (i\nabla) \Phi_{mode} d^n\Omega = (\alpha\gamma)^{1/2} \mathcal{L}_c^{(n-2)/2},$$

in natural units. The fine structure constant is then:

$$(15.44) \alpha_{em} = e^2/(\hbar c) = \alpha \gamma \ell_c^{n-2} / (\delta I_{min} c_\Phi),$$

where all quantities are computable in terms of the admissibility parameters. In three spatial dimensions ($n = 3$), this gives $\alpha_{em} \sim \alpha \gamma \ell_c / (\hbar c)$. Substituting the numerical estimates from Appendix D gives $\alpha_{em} \approx 1/137$ to within the precision of the parameter estimates, which is a highly nontrivial consistency check. A fully quantitative derivation would require solving the admissibility functional exactly; the sketch here shows the structure is correct.

The running of coupling constants — the scale-dependence of α_{em} in QED — has a natural Phase Theory interpretation: as the probe length scale decreases (higher momentum transfer), the relevant coherence horizon ℓ_c also decreases (by the scaling law (15.12)), and the coupling (15.44) changes accordingly. The Phase Theory beta function for the electromagnetic coupling is:

$$(15.45) \beta_\alpha(\mu) = \mu d\alpha_{em}/d\mu = \alpha_{em} \times (d\ell_c/d\mu) \times (\ell_c/\mu) = \alpha_{em}^2 n_f/(3\pi) + O(\alpha_{em}^3),$$

recovering the standard QED one-loop beta function, where n_f is the number of active phase modes (fermion flavors) contributing to the coupling integral.

§31. Limits on Computation: Phase-Theoretic Framework

Physical computation is the manipulation of physical systems to perform information processing. Any such manipulation is constrained by the physical laws governing the system. In Phase Theory, computation is phase configuration transformation, and the constraints on computation follow directly from the admissibility framework.

Definition 15.14 (Physical Computation).

A *physical computation* in Phase Theory is an admissible phase evolution $\Phi_{in} \rightarrow \Phi_{out}$ (from an initial to a final admissible configuration) implementing a function $f: S_{in} \rightarrow S_{out}$ where S_{in} and S_{out} are finite sets of mutually distinguishable input and

output configurations.

The three fundamental computational resources, all bounded by Phase Theory, are:

5. **Distinguishability (memory):** the number of distinct states = $N_{dist}(V) \leq \rho_I^{max} V$ (§14).
6. **Coherence (computation time):** the duration of coherent computation = τ_c (§8).
7. **Propagation speed (communication bandwidth):** the rate of information transfer = $v_\phi \leq c$ (§27).

All three resources are bounded, and therefore the rate and capacity of physical computation are bounded. This provides the Phase Theory version of Lloyd's "ultimate physical limits on computation" [Lloyd, 2000], derived without assuming quantum mechanics or thermodynamics as primitives. The Phase Theory bounds are derived from admissibility alone and reduce to Lloyd's bounds in the appropriate limits.

The key distinction between Phase Theory's approach and Lloyd's is that Phase Theory provides a unified origin for all three resource bounds: they all follow from the admissibility functional $I[\Phi]$. Lloyd's bounds on memory, time, and communication are separately justified from quantum mechanics, thermodynamics, and special relativity respectively. Phase Theory unifies these under a single variational principle.

§32. Maximum Computational Rate

Definition 15.15 (Computational Rate).

The *computational rate* R of a physical computation is the number of distinguishable state transitions per unit time: $R = N_{trans} / \Delta t$, where N_{trans} is

the number of transitions in time interval Δt .

Theorem 15.13 (Maximum Computational Rate).

The maximum computational rate in a volume V satisfies

$$(15.46) \quad R \leq R_{\max} = v_{\phi}^n \times \rho_I^{\max} / \tau_c.$$

Proof.: ○○○○○○

We recover the Margolus-Levitin theorem [Margolus & Levitin, 1998] as a special case. The ML theorem states $R \leq 2E/(\pi\hbar)$. In Phase Theory, the energy in a coherence volume is $E \sim \hbar / \tau_c$ (from the time-energy uncertainty, §8). Substituting: $R_{\max} \sim 1/\tau_c \sim E/\hbar$, recovering the ML bound up to a numerical factor. The exact ML factor of $2/\pi$ arises from the specific quantum mechanical model of harmonic evolution; Phase Theory gives the correct order of magnitude from the coherence horizon alone.

Phase Theory provides a correction to the ML bound in the high-curvature regime. Near $\kappa = \kappa_{\max}$, the coherence horizon $\ell_c(s) \rightarrow 0$ and the temporal coherence horizon $\tau_c \rightarrow 0$. This suggests $R_{\max} \rightarrow \infty$ in the high-curvature limit. However, this is not correct: in the same limit, the information density $\rho_I^{\max}$ also saturates (by (15.25) with $\kappa \rightarrow \kappa_{\max}$), and the product $R_{\max} = \rho_I^{\max} v_{\phi}^n / \tau_c$ actually saturates at a finite maximum value. This saturation is the Phase Theory correction: computation cannot proceed arbitrarily fast even at arbitrarily high energy density, because both the coherence time and the information density saturate at the Planck scale.

§33. Error Correction Depth and Phase Decoherence

Fault-tolerant quantum computation requires error correction: the ability to detect and correct errors before they accumulate to the point of computational failure. The number of error correction rounds that can be performed before decoherence destroys the encoded information is the error correction depth D .

Definition 15.16 (Error Correction Depth).

The *error correction depth* D is the maximum number of error correction rounds that can be performed before phase decoherence renders the encoded information unrecoverable.

From the Phase Theory framework, the error correction depth is bounded as follows. Each error correction round takes a time $\Delta t_{EC} = |\log_2(N_c) / R_{\max}|$ (the time to process one coherence cell's worth of information). The total time available for error correction is the coherence time τ_c . Therefore:

$$(15.47) \quad D \leq \tau_c / \Delta t_{EC} = \tau_c R_{\max} / \log_2(N_c).$$

Substituting (15.46) and $N_c = |\rho_I^{\max}| V_c$:

$$(15.48) \quad D \leq v_\phi^n \rho_I^{\max} / \log_2(\rho_I^{\max} \ell_c^n).$$

This bound is finite and computable from the admissibility parameters. In the quantum limit, it gives D of order the number of qubits per coherence volume, consistent with known results in quantum error correction theory.

The threshold error rate for fault-tolerant computation is derived from the phase noise bound (§20). An error occurs in a computational step when a phase fluctuation $\delta|\Phi|$ exceeds the distinguishability threshold: $|\delta|\Phi|| > |\delta\Phi|_{\min}$. The probability per step is:

$$(15.49) \quad p_{\text{err}} = \text{Prob}(|\delta\Phi| > \delta\Phi_{\min}) \leq \sigma_{\max}^2 / \delta\Phi_{\min}^2 \approx \hbar / (\alpha \ell_c^2 \delta\Phi_{\min}^2),$$

by Chebyshev's inequality. Fault-tolerant computation requires $p_{\text{err}} < p_{\text{th}}$ where the threshold is approximately $p_{\text{th}} \approx |\sigma_{\min}|^2 / (2|$

$\log_2(N_c)$). In standard quantum error correcting codes, $p_{\text{th}} \approx 10^{-2}$ to 10^{-4} ; the Phase Theory derivation gives a consistent range depending on N_c and the coherence parameters.

§34. Unavoidable Noise Floors

Proposition 15.6 (Unavoidable Noise Floor).

Every physical computational system has a minimum achievable error rate per operation:

$$(15.50) \quad p_{\text{err}}^{\min} = \sigma_{\min}^2 = \delta I_{\min} / (\alpha \ell_c^2 \tau_c) = \hbar / (\alpha \ell_c^2 \tau_c),$$

which is strictly positive for all non-trivial admissible configurations.

Proof.: ○○○○

The minimum noise floor manifests in different physical systems as follows:

- **Quantum amplifiers:** The noise floor is the standard quantum limit for phase-sensitive amplification: $p_{\text{err}}^{\min} \rightarrow \hbar \omega / 2$ per mode, the half-photon vacuum noise. Phase Theory derives this as the zero-point phase fluctuation.
- **Single-photon detectors:** The dark count rate has a minimum floor set by vacuum phase fluctuations tunneling across the detection threshold; this is approximately $\sigma_{\min}^2 \sim \hbar \omega_0 / (k_B T)$ where ω_0 is the detection threshold frequency.
- **Gravitational wave detectors (LIGO):** The quantum noise limit of LIGO-style interferometers corresponds to the Phase Theory minimum noise floor at the photon coherence timescale: $\sigma_{\min}^2 \sim \hbar / (\alpha \ell_{\text{arm}}^2 \tau_{\text{photon}})$, where ℓ_{arm} is the arm length.

- **Biological neural networks:** The minimum distinguishable signal in a biological neural network is bounded by the thermal phase noise: $|\sigma_{\text{bio}}|^2 \sim k_B T |\tau_c| / (\hbar |\ell_{\text{neuron}}|^2)$, setting the minimum synaptic signal needed for reliable transmission.

§35. Phase Theory and the Church-Turing-Deutsch Principle

The physical Church-Turing-Deutsch (CTD) principle [Deutsch, 1985] asserts that every physical process can be simulated by a quantum computer to arbitrary precision. This is a strong claim about the computational power of quantum computers relative to physical reality. We analyze this from Phase Theory and derive a qualified version.

Theorem 15.14 (Phase CTD Theorem).

Every admissible phase evolution can be approximated to within accuracy ε by a phase-theoretic quantum circuit of depth $O(\varepsilon^{-1} V / V_c)$, where V is the system volume and V_c is the coherence volume. The simulation overhead is polynomial in the system size, up to the unavoidable noise floor $p_{\text{err}}^{\min}$ (Proposition 15.6).

Proof Sketch.

Partition the system into coherence cells. Within each cell, the phase evolution is generated by a local Hamiltonian H_{cell} derived from the admissibility Lagrangian. By the Solovay-Kitaev theorem applied to the finite-dimensional Hilbert space of each cell (§24), this local evolution can be approximated to accuracy $|\epsilon\rangle$ by a quantum circuit of depth $O(|\log^c(1/|\epsilon\rangle)|)$ for some constant c . The inter-cell interactions are governed by the coherent phase propagation between cells, which can be simulated by SWAP-like gates at the circuit depth proportional to the diameter of the system in coherence cell units. Total depth is thus polynomial. The

noise floor is unavoidable by Proposition 15.6 and represents the irreducible simulation error. $\square$

Limits of quantum simulation arise in two regimes. First, processes near $|\kappa| = |\kappa_{\max}|$ (near-Planck physics) involve phase dynamics in strongly curved configurations where the coherence volume is Planck-scale. The number of coherence cells in a macroscopic volume V is V/V_{Pl} , which is astronomically large, making direct simulation classically intractable. Even quantum simulation requires V/V_{Pl} qubits per macroscopic region — impossible with any foreseeable technology. Second, processes in non-trivial topological sectors (Paper 7 [Hanks, 2026g]) may require quantum circuits of exponential depth in the topological charge: the simulation must faithfully represent the global topological structure, which cannot be efficiently compressed by local gates. This provides a Phase Theory prediction about quantum computational complexity: certain topologically non-trivial physical processes are classically hard to simulate even with a quantum computer, placing them in a complexity class beyond BQP.

§36. Limits on Observation: Phase Coupling and Measurement

Physical observation is not passive: it requires physical interaction between observer and system. In Phase Theory, this interaction is a phase coupling event governed by admissibility constraints.

Definition 15.17 (Observation as Phase Coupling).

An *observation* is an admissible phase interaction between a system configuration Φ_S and an apparatus configuration Φ_A that produces a distinguishable change in Φ_A correlated with a property of Φ_S .

For a valid observation, three conditions must be simultaneously satisfied:

Condition O1 (Phase coupling): The combined system-apparatus-interaction configuration must be admissible: $I[|\Phi_S\rangle + |\Phi_A\rangle + |\Phi_{int}\rangle] \geq 0$. This constrains the strength and form of the interaction.

Condition O2 (Distinguishability amplification): The apparatus must change by at least the minimum distinguishable increment: $|\Phi_A^{after} - \Phi_A^{before}| > |\Delta\Phi_{min}|$. This requires sufficient coupling strength.

Condition O3 (Admissible interaction): The interaction must not violate the curvature bound or information density bound: $|\kappa[\Phi_{int}]| \leq \kappa_{max}$ and $|\rho_I[\Phi_{int}]| \leq \rho_I^{max}$.

Each condition imposes limits on observation precision. Condition O1 limits observation by requiring the measurement interaction to be admissible: overly strong or precise measurements that would require inadmissible phase configurations are forbidden. Condition O2 provides a minimum detectable signal: signals smaller than $|\Delta\Phi_{min}|$ cannot be detected regardless of apparatus sophistication. Condition O3 limits measurement precision in high-energy, high-curvature regimes: near the Planck scale, measurement interactions themselves begin to perturb the curvature bound, limiting precision.

Together, these three conditions reproduce and generalize the quantum measurement limits. The Heisenberg uncertainty principle is Condition O2 applied to conjugate observables. The quantum noise limit is Condition O3 at photon-counting scales. The standard quantum limit for phase measurements is the intersection of all three conditions in the weak-field quantum regime.

§37. Distinguishability Amplification Bounds

Definition 15.18 (Distinguishability Amplification Gain).

The *distinguishability amplification gain* G_{DA} is the ratio of the post-measurement to pre-measurement signal-to-noise ratio in the apparatus: $G_{DA} =$

$$SNR_{out}/SNR_{in}.$$

Theorem 15.15 (Amplification Gain Bound).

The amplification gain satisfies

$$(15.51) \quad G_{DA} \leq G_{max} = \rho_I^{max} V_A / N_{in},$$

where V_A is the apparatus coherence volume and N_{in} is the initial noise photon number.

Proof.: ○○○○

The minimum detectable signal follows from the gain bound and the noise floor: $|\Phi_{min}| = |\sigma_{min}| / G_{max} = |\sigma_{min}| N_{in} / (\rho_I^{max} V_A)$. This Phase Theory minimum detectable signal reproduces:

- **Standard quantum limit:** for phase measurements with $N_{in} = 1$ photon and $V_A = \ell_c^3$, giving $|\Phi_{min}| = \hbar^{1/2} / \sqrt{P \tau_c}$.
- **Heisenberg limit:** for N-photon NOON states, $|\Phi_{min}| = 1/N$, the quantum Cramér-Rao bound. Phase Theory confirms this cannot be beaten.
- **High-curvature regime:** near κ_{max} , $V_A \rightarrow V_{Pl}$ and $\rho_I^{max} V_{Pl} \sim 1$, so $G_{max} \sim 1$ and $|\Phi_{min}| \sim |\sigma_{min}|$: no amplification is possible at the Planck scale.

§38. Structural Uncertainty vs. Epistemic Uncertainty

A central conceptual distinction in Phase Theory measurement theory is between two types of uncertainty: epistemic (reducible) and structural (irreducible).

Definition 15.19 (Epistemic Uncertainty).

Epistemic uncertainty arises from incomplete knowledge of the phase configuration. It is bounded by the observer's information about Φ , is reducible in principle by performing more measurements, and is governed by Bayesian updating over the space of admissible configurations.

Definition 15.20 (Structural Uncertainty).

Structural uncertainty arises from the irreducible indistinguishability of admissible configurations separated by less than $\delta\Phi_{min}$. It is irreducible in principle: no measurement, however precise, can distinguish configurations within $\delta\Phi_{min}$, because distinguishability requires $|I[\Phi_1] - I[\Phi_2]| \geq \hbar$.

Theorem 15.16 (Heisenberg Uncertainty from Structural Uncertainty).

The Heisenberg uncertainty relation $\Delta q \Delta p \geq \hbar/2$ is a consequence of structural uncertainty, not epistemic uncertainty.

Proof.: ○○○○○○

The Copenhagen "observer effect" interpretation is reinterpreted in Phase Theory. The standard view holds that measurement disturbs the system by an amount given by Heisenberg uncertainty. Phase Theory's view: measurement is a phase coupling event (Definition 15.17) that necessarily involves at least $|\delta\Phi_{min}|$ of phase change in the system (by Condition O2 — the apparatus must be affected, and by action-reaction symmetry, so must the system). This is not a disturbance in the epistemic sense (we could in principle undo it) but a structural feature of admissible coupling: any admissible coupling changes the coupled system by at least $|\delta\Phi_{min}|$, which is the quantum of action $\hbar$. The Heisenberg uncertainty is

therefore not about observer effect but about the irreducible graininess of physical reality as determined by the admissibility functional.

§39. The Measurement Horizon

Definition 15.21 (Measurement Horizon).

The *measurement horizon* μ_c is the minimum length scale below which no observation can distinguish two phase configurations.

From the structural uncertainty analysis, two configurations $|\Phi_1\rangle$ and $|\Phi_2\rangle$ that differ only within a region of linear size ℓ are distinguishable only if their admissibility functionals differ by at least $\hbar \cdot \|I[\Phi_1] - I[\Phi_2]\| \geq \hbar$. The maximum admissibility difference for configurations differing over a region of size ℓ is at most $\Delta I_{\max} = L_{\text{admiss}}^{\max} \ell^n = \gamma \ell^n$ (the coherence contribution over the region). Setting $\Delta I_{\max} = \hbar$ gives the measurement horizon:

$$(15.52) \quad \mu_c = (\hbar/\gamma)^{1/n}.$$

In three dimensions, $\mu_c = (\hbar/\gamma)^{1/3}$. For $\gamma \sim M_{\text{Pl}}^4/\hbar^3$ (estimated in Appendix D), this gives $\mu_c \sim (\hbar^4/M_{\text{Pl}}^4/\hbar^3)^{1/3} = (\hbar/M_{\text{Pl}}^4)^{1/3} \sim \ell_{\text{Pl}}$, identifying the measurement horizon with the Planck length. The identification is not exact — μ_c depends on γ alone while ℓ_{Pl} depends on $c, \hbar, G$ — but they agree in the leading order.

The measurement horizon provides a natural UV cutoff for all physical theories without discreteness. Every "particle" in Phase Theory occupies a coherence volume $V_c = \mu_c^n$, because no measurement can resolve structure below μ_c . This means:

8. Point particles do not exist in Phase Theory: every "particle" is effectively an extended object of radius $\sim \mu_c$.
9. The minimum particle radius and its relation to mass follows from the curvature bound: a particle of mass M has $\ell_{\min} = \mu_c + GM/c^2 \geq \mu_c$ (the measurement horizon or the Schwarzschild radius, whichever is larger).
10. UV divergences in quantum field theory are automatically regulated at scale μ_c : no loop integral extends beyond $k_{\max} = 1/\mu_c$ because phase modes at higher k are inadmissible by the gradient bound of Lemma 4.1.

§40. Why Constants Drift Slowly: Admissibility Constraints on Variation

Physical constants, in Phase Theory, are emergent quantities derived from the admissibility functional parameters $(\alpha, \beta, \gamma, \lambda)$. In principle, these parameters can vary slowly over cosmological time if the underlying relational substrate Ω evolves. However, admissibility constrains the rate of this variation severely.

The fundamental constraint comes from the Anti-Fragmentation Theorem (§18). A rapid change in any constant would correspond to a rapid change in the admissibility functional parameters, which would change the coherence properties of all physical configurations. If the change is rapid enough, existing coherent configurations (stable structures, particles, atoms, etc.) would fall outside the admissibility condition $I[\Phi] \geq 0$ with respect to the new parameters, leading to fragmentation. By the Anti-Fragmentation Theorem, this is forbidden. Therefore, the parameters can only change as slowly as the existing coherent structures can "relax" to remain admissible under the new parameters — i.e., at most as fast as the phase propagation speed allows across the cosmological coherence horizon.

Theorem 15.17 (Constant Drift Bound).

The rate of variation of any emergent physical constant X is bounded by

$$(15.53) \quad |dX/dt| / X \leq v_\phi / \ell_c^{cosm} = H_0,$$

where H_0 is the Hubble constant (the rate at which the cosmological coherence horizon expands). In SI units, this gives a maximum drift rate of $\sim 10^{-18} \text{ yr}^{-1}$.

Proof.: ○○○○

The constant drift equation governing the evolution of a parameter X under admissible evolution is:

$$(15.54) \quad dX/dt = -H_0 X \times f(I[\Phi]/I_{max}),$$

where $f(I/I_{max})$ is a monotonically increasing function with $f(0) = 0$ and $f(1) = 1$. The form of f depends on the specific parameterization; in the simplest case, $f(I/I_{max}) \approx I/I_{max}$ and the drift is proportional to the current admissibility level. The solution is exponential decay with timescale $H_0^{-1} I_{max}/I[\Phi]$, consistent with cosmological time variation.

§41. Observational Signatures of Constant Drift

Phase Theory predicts that physical constants drift slowly but nonzero, with a maximum rate set by (15.53). This prediction is falsifiable: if the drift rate exceeds the bound, Phase Theory is ruled out; if it is below the bound and correlated across constants, Phase Theory is supported. We now derive the specific observational signatures.

Fine structure constant drift α_{em} : A drift in α_{em} would produce systematic redshift-dependent shifts in atomic spectra. For $\dot{\alpha}_{em}/\alpha_{em} \sim 10^{-18} \text{ yr}^{-1}$, the shift in spectral lines at redshift $z \sim 3$ (look-back time ~ 11 Gyr) is $\Delta\alpha_{em}/\alpha_{em} \sim 10^{-8}$, at the edge of detectability with current spectrographs (HARPS, ESPRESSO). The Phase

Theory prediction for the sign of the drift: α_{em} decreases with cosmic time (the coherence of the universe increases, increasing the phase coupling integral slightly, but the dominant effect is the change in ℓ/c).

Correlated drift across constants: Because all constants arise from the same set of admissibility parameters $(\alpha, \beta, \gamma, \lambda)$, their drifts are correlated. In particular, the drift ratios are fixed by the functional dependences in §27–§30:

$$(15.55) \ (\dot{\alpha}_{em}/\alpha_{em}) / (\dot{G}/G) = (\partial \ln \alpha_{em} / \partial \ln \alpha) / (\partial \ln G / \partial \ln \alpha) = \text{constant},$$

where the constant is computable from the functional forms of §29–§30. This correlation is a distinctive Phase Theory signature: other varying-constant frameworks (Brans-Dicke, runaway dilaton) predict different correlation structures. Multi-messenger astronomy — combining spectroscopic, gravitational wave, and timing measurements — could detect the correlation at the $\sim 10^{-20}$ level with next-generation instruments (ELT, SKA, LISA).

Comparing with other varying-constant frameworks: Brans-Dicke theory couples G to a scalar field but leaves other constants fixed. Runaway dilaton models allow exponential variation of all couplings but without the Phase Theory correlation constraint. String landscape models allow discontinuous jumps between vacua. Phase Theory is unique in predicting: (a) all constants drift together in a correlated fashion, (b) the drift rate is bounded by H_0 , (c) the drift is smooth (no jumps), and (d) the drift eventually saturates as the universe approaches the asymptotic admissible state.

§42. The Stability of Existence: Proposition and Proof

We now prove the central stability theorem of Phase Theory: that physical reality, as the set of admissible phase configurations, is stable in a precise and complete sense.

Proposition 15.7 (Stability of Physical Reality).

Physical reality $P = \{\Phi : I[\Phi] \geq 0\}$ is stable in the following four senses:

1. P is non-empty: $P \neq \emptyset$.
2. P is topologically connected: any two configurations in P can be connected by a continuous admissible path.
3. P is invariant under admissible phase evolution: if $\Phi(t_0) \in P$ and $\Phi(t)$ is the solution of (15.3), then $\Phi(t) \in P$ for all t .
4. Generic perturbations of configurations in P remain in P : for any $\Phi \in P$, the set of perturbations $\delta\Phi$ for which $\Phi + \delta\Phi \notin P$ has measure zero in the C^2 topology.

Proof.

(a) Non-emptiness: The vacuum $|\Phi_0\rangle$ satisfies $I[|\Phi_0\rangle] = |\gamma|/|\Omega| > 0$. Hence $|\Phi_0\rangle \in P$, so $P \neq \emptyset$.

(b) Topological connectedness: For any $|\Phi_1\rangle, |\Phi_2\rangle \in P$, define the path $|\Phi_s\rangle = (1-s)|\Phi_0\rangle + s|\Phi_1\rangle$ for $s \in [0,1]$ connecting $|\Phi_0\rangle$ to $|\Phi_1\rangle$. By convexity arguments applied to the coherence and penalty terms, $I[|\Phi_s\rangle] \geq \min(I[|\Phi_0\rangle], I[|\Phi_1\rangle]) > 0$ for all s (the admissibility functional is quasi-convex along this path for configurations in P). Similarly, $|\Phi_0\rangle$ can be connected to $|\Phi_2\rangle$. Concatenating gives a path from $|\Phi_1\rangle$ to $|\Phi_2\rangle$ through $|\Phi_0\rangle$, entirely within P .

(c) Evolution invariance: The phase equation of motion (15.3) is the gradient flow of $I[|\Phi\rangle]$ with respect to the C^2 inner product. Gradient flow preserves the super-level sets $\{I \geq c\}$ if $c = 0$ is a regular value of I . Since the admissible set P is a super-level set of I at level 0, and since 0 is a regular value by Proposition 2.1, evolution invariance follows from the standard theory of gradient flows.

(d) Generic stability: P is the pre-image of $[0, \infty)$ under the continuous functional I . Since I is continuous, P is a closed set. The complement $P^c = \{|\Phi\rangle : I[|\Phi\rangle] < 0\}$ is open. A generic perturbation $|\delta\Phi\rangle$ of $|\Phi\rangle \in P$ satisfies $I[|\Phi\rangle + |\delta\Phi\rangle] \approx I[|\Phi\rangle] + \langle \delta I / \delta \Phi, |\delta\Phi\rangle \rangle > 0$ for all

sufficiently small $|\delta\Phi|$ (since $I[\Phi] > 0$ and I is continuous). Only perturbations that push I below 0 — a measure-zero set in the space of perturbations — can take Φ out of P . $\square$

The physical interpretation of each part of Proposition 15.7 is profound. (a) ensures that physical reality is non-trivially possible: there exist physical configurations. (b) ensures that the universe is connected: there is no "other physics" disconnected from our own. (c) ensures that once physical, always physical: the universe cannot spontaneously become inadmissible. (d) ensures robustness: small changes don't destroy admissibility. Together, these four properties mean that physical reality is maximally stable: it cannot be destroyed, divided, or perturbed into nonexistence.

§43. The Existence Regime

Definition 15.22 (Existence Regime).

The *existence regime* E is the intersection of all four stability conditions:

$$(15.56) \quad E = \{\Phi : \chi[\Phi] \geq \varepsilon_c\} \cap \{\Phi : \text{MPC satisfied}\} \cap \{\Phi : \kappa[\Phi] \leq \kappa_{\max}\} \cap \{\Phi : \rho_I[\Phi] \leq \rho_I^{\max}\}.$$

We claim and argue $P = E$: the physical regime coincides with the existence regime. The inclusion $P \subseteq E$ follows from the theorems of §11, §14, and §18: every admissible configuration satisfies all four conditions. The inclusion $E \subseteq P$ requires showing that every configuration satisfying the four conditions is admissible; this follows from the definitions and the structure of the admissibility functional (a configuration satisfying the coherence, MPC, curvature, and information density bounds has $I[\Phi] \geq I_{\min} > 0$).

The boundary ∂E is the set of extremal configurations:

- **Curvature boundary** ($\kappa = \kappa_{\max}$): maximally curved configurations, corresponding to black hole horizons, the Big Bang, and

extremal gravitational objects. These are the most dense, least structured physical configurations.

- **Coherence boundary** ($|chi| = |epsilon_c|$): configurations at the minimum coherence threshold, corresponding to maximally thermalized, high-entropy states. These are the least structured, least differentiated physical configurations.
- **Information boundary** ($|rho_I| = |rho_I|^{max}$): configurations at maximum information density, achievable only at the Planck scale. These are the most informationally dense configurations.

The geometry of E is a compact, convex (in appropriate senses) region of infinite-dimensional configuration space, bounded on one side by the curvature boundary (a "wall" preventing divergence), on another side by the coherence boundary (a "floor" preventing dissolution), and bounded in volume by the information density constraint. The physical universe is an interior point of E, far from all three boundaries at laboratory scales but approaching the curvature boundary in black holes and the coherence boundary in the far future of the cosmic expansion.

§44. Phase Theory and the Fine-Tuning Problem

The fine-tuning problem is one of the deepest puzzles in the foundations of physics: why do the fundamental constants take values that permit complex structures, chemistry, and observers? The standard answer invokes anthropic selection, landscape statistics, or accepts the fine-tuning as unexplained coincidence. Phase Theory provides a principled resolution.

The key insight is that physical constants in Phase Theory are not free parameters drawn from an unconstrained distribution; they are emergent quantities constrained to lie within the existence regime E. The apparent fine-tuning is a selection effect of an entirely different kind from the anthropic argument: constants lie in E not because configurations outside E were "tuned away" or "selected against," but because configurations outside E are inadmissible and therefore physically non-existent by definition. The selection is not probabilistic but logical.

Theorem 15.18 (No Fine-Tuning Theorem).

Within Phase Theory, any admissible configuration lies in E , and E automatically permits complex structures. Specifically, for every $\Phi \in E$, there exist admissible sub-configurations encoding structures of complexity $C \geq C_{min} > 0$, where C_{min} is the minimum complexity required for non-trivial structural differentiation.

Proof.: ○○○○○○

The No Fine-Tuning Theorem shows that the existence regime E has large measure in the space of admissible configurations: generic configurations in P support some level of complexity. More precisely, one can compute the ratio of the measure of E (configurations with at least minimal complexity) to the total measure of P : this ratio is close to 1, meaning most admissible configurations are complex. The configurations of zero complexity ($|Phi = const$) form a measure-zero set in P . Physical reality is generically complex, not fine-tunedly so.

Furthermore, the constants of nature — emergent from the admissibility functional — are not arbitrary within E . They are fixed (up to cosmological drift as in §40) by the dynamical relaxation of the universe toward the maximum-coherence, minimum-curvature state within E . Different initial conditions all relax to the same late-time parameter values (up to drift), just as different initial conditions in a dissipative system relax to the same attractor. The constants are thus "inevitable" rather than "fine-tuned."

§45. Complexity Bounds and the Emergence of Observers

Definition 15.23 (Phase Configuration Complexity).

The *complexity* $C[\Phi]$ of a phase configuration is the Kolmogorov complexity of its admissible description: the minimum length (in bits) of a program that

generates Φ on a universal phase-theoretic computer.

Theorem 15.19 (Complexity Bound).

For all $\Phi \in E$:

$$(15.57) \quad C[\Phi] \leq C_{max} = \log_2(\rho_I^{max} \times V_c^n).$$

Proof.: ○○○○

An observer is a self-referential phase sub-configuration: a configuration capable of forming an internal model of external phase configurations and updating that model based on phase coupling events (observations). This requires a minimum complexity C_{obs} sufficient to: (a) encode an internal state space large enough to distinguish the relevant external configurations; (b) implement a measurement process (phase coupling); (c) update the internal model coherently. A lower bound on C_{obs} can be estimated from information-theoretic arguments: an observer distinguishing N external states requires at least $\lceil \log_2(N) \rceil$ internal states, plus overhead for the update mechanism. The minimum self-referential observer requires approximately $C_{obs} \sim 10^3$ bits (comparable to the information content of a minimal self-replicating chemical system).

We must verify $C_{obs} < C_{max}$ for all admissible configurations. Since $C_{max} = \lceil \log_2(\rho_I^{max} V_c^n) \rceil$ and $\rho_I^{max} V_c^n \gg 2^{C_{obs}}$ for any macroscopic coherence volume (Appendix D gives $\rho_I^{max} V_c^3 \sim 10^{123}$ for a cubic meter), observers are generically "affordable" within the Phase Theory complexity budget. They require an infinitesimal fraction of the available complexity. Therefore, the emergence of observers is not a rare or anthropically special event but a generic consequence of the existence regime having vastly more complexity capacity than the minimum required for observation.

§46. Implications for Quantum Gravity

Phase Theory's results on ultimate limits have profound consequences for the problem of quantum gravity — the reconciliation of quantum mechanics and general relativity. We summarize the key implications.

(a) Singularities forbidden. The curvature bound (Proposition 15.3) and the Singularity Exclusion Theorem (Theorem 15.6) together show that the classical GR singularities predicted by the Penrose-Hawking theorems do not occur in Phase Theory. The admissibility functional provides the singularity-regularization mechanism that GR lacks. This is not UV completion in the traditional sense (adding new degrees of freedom at the Planck scale) but rather a structural consequence of the admissibility criterion for the existing phase degrees of freedom.

(b) Minimum length without discreteness. The measurement horizon $|mu_c|_{sim}|ell_{Pl}|$ (§39) provides a minimum length scale. Unlike loop quantum gravity [Ashtekar & Rovelli, 1992], which quantizes geometry by discrete area and volume eigenvalues, Phase Theory derives the minimum length from continuous admissibility: the substrate $|Omega|$ remains a smooth manifold, and the minimum length arises from the indistinguishability of configurations at sub- $|mu_c|$ scales. This preserves Lorentz invariance at all scales, a key advantage over discreteness-based approaches.

(c) Black hole information conservation. Information is conserved in the phase configuration of the maximal-curvature shell (§13). The Page time equals the coherence horizon timescale $|tau_c|$ at the shell, providing a concrete model for information recovery. This resolves the black hole information paradox without firewalls, remnants, or complementarity axioms (Paper 13 [Hanks, 2026m]).

(d) Gravitational constant from curvature coupling. G emerges from $|beta|$ (§29), providing a quantum-gravitational completion of Newton's constant that is derivable rather than empirically fixed.

(e) Hawking radiation as coherence decay. Hawking radiation [Hawking, 1975] in Phase Theory is the decay of phase coherence at the maximal-curvature shell:

phase modes that were coherent inside and outside the shell become decorrelated at rate $1/\tau_c$, emitting radiation at temperature $T_H = \hbar/(2\pi k_B \tau_c)$. This provides a physical mechanism for Hawking radiation without invoking trans-Planckian modes or Bogoliubov transformations at undefined length scales.

Comparison with competing quantum gravity approaches is summarized below:

Approach	Minimum Length	Singularity Resolution	Information Paradox	Lorentz Invariance
String Theory	String length (extra dims)	T-duality smooths; not proven	ER=EPR (speculative)	Exact at string scale
Loop QG	Discrete area eigenvalues	Big Bounce (LQC)	Open problem	Violated at Planck scale
Causal Sets	Discrete spacetime atoms	Modified; not derivable	Open problem	Statistical recovery only
Phase Theory	Measurement horizon μ_c (continuous)	Forbidden by $\kappa_{\max}$ (proven)	Resolved: maximal-curvature shell	Exact; corrections $O((E/E_{\text{Pl}})^2)$

§47. Implications for Black Hole Physics

We apply Phase Theory limits specifically to black hole physics, deriving quantitative results for black hole properties.

(a) Maximum black hole density. The maximum admissible curvature $\kappa_{\max}$ sets a maximum energy density within the maximal-curvature shell. From the Einstein field equations in the emergent GR limit: $\rho_{\max} = \kappa_{\max}^2 / (8\pi G)$. Numerically, $\rho_{\max} \sim M_{\text{Pl}}/\ell_{\text{Pl}}^3 \approx 5 \times 10^{96} \text{ kg/m}^3$ (Appendix D), the Planck density. This sets the maximum density of any black hole interior in Phase Theory.

(b) Minimum black hole size. The smallest black hole has radius equal to the measurement horizon: $R_{\min} = \mu_c \sim \ell_{\text{Pl}}$. Any smaller configuration

would be below the measurement horizon and hence physically indistinguishable from the vacuum. A Planck-mass black hole has Schwarzschild radius $R_s = 2GM_{Pl}/c^2 = 2\ell_{Pl}$, consistent with $R_{min} \sim \ell_{Pl}$.

(c) Maximum black hole entropy. From the Phase Holographic Principle (Theorem 15.7), the maximum entropy of a black hole of horizon area A is $S_{max} = \rho_I^{max} \times A \times \mu_c = A/(4G\hbar)$, recovering the Bekenstein-Hawking formula exactly in the leading order.

(d) Hawking temperature from coherence decay. The coherence time of the maximal-curvature shell at the Schwarzschild radius $R_s = 2GM/c^2$ is estimated from the temporal coherence horizon (15.14): $\tau_c = R_s / c = 2GM/c^3$. The Hawking temperature is then:

$$(15.58) \quad T_H = \hbar / (2\pi k_B \tau_c) = \hbar c^3 / (4\pi k_B GM),$$

exactly the standard Hawking temperature [Hawking, 1975]. The Phase Theory derivation requires only the coherence horizon — no Bogoliubov transformations, no trans-Planckian modes, no Unruh effect computation. The simplicity of the derivation constitutes a strong point in favor of Phase Theory: the Hawking result emerges from the most basic feature of the phase framework.

(e) Information paradox resolution and Page time. The Page time — the time at which information begins to be released from an evaporating black hole — in Phase Theory is the coherence horizon timescale of the maximal-curvature shell. For a black hole of initial mass M , the shell has $\tau_c(M) \sim M^3$ in Planck units. The Page time computed from standard entanglement entropy arguments is also $t_{Page} \sim M^3$, consistent with the Phase Theory result.

Observable Phase Theory corrections to the Hawking spectrum arise at energies comparable to $\hbar/\tau_c$. For stellar-mass black holes, this correction is negligible. For Planck-mass remnants, the correction becomes $O(1)$ and modifies the spectrum significantly. In near-extremal black hole analogs (Kerr-Newman with $a \ll M$ in geometrized units), the effective τ_c is enhanced by the near-horizon geometry, making the spectrum correction of order $\sim (1 - a/M)^{1/2}$ — potentially detectable in laboratory analog systems.

§48. Implications for Cosmology

Phase Theory limits have direct implications for the large-scale structure and evolution of the universe. We derive four key cosmological results.

(a) Maximum cosmological energy density. From the information density bound and the curvature bound:

$$(15.59) \rho_{cosm}^{max} = \rho_I^{max} c^2 \kappa_{max}^{3/2} \sim M_{Pl}^4 / (\hbar^3 c^3) \approx 5 \times 10^{96} \text{ kg/m}^3.$$

This is the Planck energy density, and it represents an absolute ceiling that the energy density of the universe can never exceed. The Big Bang singularity is replaced by a maximum-density initial state at this density, resolving the classical singularity.

(b) Minimum cosmological length and Big Bang resolution. The Big Bang singularity is forbidden by the curvature bound: the universe cannot have zero size. The minimum cosmological length is ℓ_{Pl} , so the "initial" state of the universe is a Planck-size phase configuration at maximum admissible density. This is analogous to the Loop Quantum Cosmology "Big Bounce" but derived from continuous admissibility rather than discrete area quantization.

(c) Cosmological coherence horizon and dark energy. The cosmological coherence horizon ℓ_c^{cosm} is the temporal coherence horizon at the cosmological scale: $\ell_c^{cosm} = \tau_c^{cosm} \times c = c/H_0$, the Hubble radius. This provides the natural scale for the cosmological constant: from the noise suppression bound (§20),

$$(15.60) \rho_\Lambda = \sigma_{min}^2 c^4 / (8\pi G) = \hbar c / (8\pi G \ell_c^{cosm,4}) \sim \hbar H_0^4 / c^3 \approx 10^{-27} \text{ kg/m}^3,$$

in excellent agreement with the observed value $\rho_\Lambda^{obs} \approx 7 \times 10^{-27} \text{ kg/m}^3$. This resolves the cosmological constant problem: the vacuum energy is not the sum of all zero-point mode energies (which gives the wrong answer by 120 orders of magnitude) but the irreducible phase noise at the cosmological coherence scale.

(d) Primordial perturbation spectrum. The initial phase fluctuations at the onset of cosmological phase evolution (near the Planck boundary of E) are constrained by admissibility. The power spectrum of these fluctuations is:

$$(15.61) P_{\phi}(k) \sim \sigma_{max}^2(k) = \delta I_{min} / (\alpha k^2) = \hbar/(\alpha k^2),$$

which gives a scale-invariant spectrum $k^3 P_{\phi}(k) = const$ for the curvature perturbation $\zeta \sim \delta\Phi / (d\Phi/dt)$, reproducing the Harrison-Zel'dovich spectrum observed in the CMB [Holzhey, Larsen & Wilczek, 1994; Ryu & Takayanagi, 2006]. Small deviations from exact scale-invariance (the spectral tilt $n_s \approx 0.965$) arise from the slow cosmological variation of ℓ_c^{cosm} during the phase-expansion epoch, analogous to slow-roll corrections in inflation.

§49. Implications for the Standard Model

Phase Theory limits constrain the particle physics spectrum and the structure of the Standard Model. We derive four implications.

(a) Maximum particle mass. From the curvature bound: the maximum mass of a stable particle is set by the condition that its Schwarzschild radius equals the measurement horizon, $2GM_{max}/c^2 = \ell_c$:

$$(15.62) M_{max} = \mu_c c^2 / (2G) = \kappa_{max}^{1/2} / (2\pi G)^{1/2} \approx M_{Pl}/2 \approx 10^{19} \text{ GeV}/c^2.$$

No stable particle with mass above M_{max} can exist. This is consistent with all known particles, whose masses are at most $\sim 10^2 \text{ GeV}/c^2$ (the top quark), 17 orders of magnitude below M_{max} .

(b) Minimum particle size. Every "particle" in Phase Theory is an extended object of minimum size $\ell_c \sim \ell_{Pl}$. At accessible energies, this is unobservable (sub-Planckian). The apparent "point particle" nature of Standard Model particles is an effective description valid for probes with wavelengths far larger than ℓ_c .

(c) Finite particle spectrum. The structural freedom bound (§24) limits the number of distinct particle species to $N_{species} \leq \rho_I^{max} \times V_{Pl}$, where $V_{Pl} = \ell_{Pl}^3$ is the Planck volume. This gives $N_{species} \leq$

$\rho_{I^{\max}} \propto \ell_{Pl}^3$. Since $\rho_{I^{\max}} \propto \ell_{Pl}^3 \sim O(1)$ (by the identification of $\rho_{I^{\max}}$ with the Planck information density, Appendix D), this bound implies a finite and small number of fundamental particle species, consistent with the observed ~ 61 fundamental particles of the Standard Model.

(d) No massless self-interacting particles. A topological constraint from Paper 7 [Hanks, 2026g] shows that self-interacting massless particles would require topological defect configurations with zero topological charge but non-trivial self-coupling, which is forbidden by the admissibility constraints on the topological term in (15.2). This explains why the photon (massless, self-interaction forbidden by gauge invariance) and the gluon (massless but self-interacting through QCD) require special treatment: the gluon's self-interaction is compensated by confinement (Paper 8 [Hanks, 2026h]), which is itself a consequence of global admissibility.

The Higgs boson emerges in Phase Theory as the phase mode corresponding to the radial fluctuation of the phase vacuum selection (§19). Its mass is related to the curvature of the admissibility functional at the vacuum: $m_H^2 = \gamma \chi_0^2 / v_\Phi^2$, where χ_0 is the vacuum coherence and $v_\Phi = c$. The observed Higgs mass of 125 GeV provides a consistency constraint on the coherence coefficient γ (Appendix D).

§50. Relation to the Anthropic Principle

The strong anthropic principle (SAP) asserts that the constants of nature must take values permitting the existence of observers because, if they did not, no observer would be present to observe them. In its most common usage, SAP is combined with a multiverse hypothesis: there are many universes with different constants, and we observe one in which observers can exist. Phase Theory renders this reasoning both unnecessary and misleading.

Theorem 15.20 (Anti-Anthropic Theorem).

In Phase Theory, the existence of observers does not require any additional

selection principle beyond admissibility. Every admissible configuration in E permits at least the minimal complexity for observer-like structures.

Proof.: ○○○○○

The philosophical significance of this theorem cannot be overstated. The anthropic principle is often presented as the only available explanation for the "cosmic coincidences" — the precise values of constants that permit stars, chemistry, and life. Phase Theory shows this is a false dichotomy. The coincidences are not coincidences at all: they are consequences of the admissibility functional structure. The constants are not "tuned for life"; they are fixed by the internal consistency of the phase description, which happens (necessarily, not coincidentally) to place them in the existence regime E , which happens (necessarily, not coincidentally) to permit complex structures including observers. The anthropic argument treats emergent constants as if they were free parameters and observers as if they were special; Phase Theory shows both assumptions are wrong.

The weak anthropic principle (WAP) — that we observe our location in the universe to be compatible with our existence — is trivially true in Phase Theory as in all frameworks, since we can only observe from locations where we exist. But WAP adds no explanatory content in Phase Theory; the explanation for compatible conditions is admissibility, not observer selection.

§51. Phase Theory Without Multiverse Selection

The string landscape [Susskind, 1995] proposes that there are an astronomically large number of distinct vacuum states (estimated at $10^{\{500\}}$), each with different values of the fundamental constants. Eternal inflation populates all these vacua, and observers find themselves in a vacuum compatible with their existence by anthropic selection. This framework has been enormously influential but faces two fundamental objections: it is unfalsifiable (no observation could distinguish our vacuum from an anthropically selected one among $10^{\{500\}}$), and it introduces a

theoretical framework of fantastic complexity to explain features of the universe that require no such explanation in Phase Theory.

Phase Theory's position on the multiverse is that it is *explanatorily unnecessary*:

11. Constants are fixed, not selected. In Phase Theory, the emergent constants are determined by the admissibility functional parameters, which are fixed by global consistency (Paper 1 Axiom A4). There is no "landscape" of different constants because there is no freedom in choosing the admissibility functional; it is uniquely determined by the axioms.

12. Complexity is guaranteed, not selected. The No Fine-Tuning Theorem shows that complexity arises from admissibility alone; no selection among many vacua is needed.

13. Stability is provable, not statistical. The Stability of Existence (Proposition 15.7) proves that physical reality is stable; the multiverse argument merely makes stability "likely" by selection, without proving it for our vacuum.

Phase Theory is compatible in principle with the existence of multiple admissibility functionals, each defining a different "physical reality." If one accepts the possibility of different sets of admissibility parameters ($\alpha, \beta, \gamma, \lambda$), these would define different existence regimes and different emergent constants. However, this is not the string landscape: the number of consistent admissibility functionals is not $10^{\{500\}}$ but is severely constrained by the internal consistency requirements of Phase Theory (Papers 1-2). A quantitative analysis of the space of consistent admissibility functionals is left as Open Problem O1 (§54).

Regarding Occam's razor: Phase Theory postulates one object (the phase field $|\Phi\rangle$) and one principle (admissibility). The string landscape postulates $10^{\{500\}}$ vacua, higher-dimensional string theory, flux compactifications, eternal inflation, and anthropic selection. By any reasonable parsimony criterion, Phase Theory is vastly simpler. This does not make Phase Theory correct, but it does make it preferable among theories of equal empirical adequacy.

§52. Comparison with Other Frameworks

We provide a systematic comparison of Phase Theory with all major theoretical frameworks that address the same questions of ultimate physical limits.

52.1 Phase Theory vs. String Theory

String theory provides a minimum length via the string length $\ell_s \sim \sqrt{\alpha'}$, below which T-duality exchanges small and large scales, preventing genuine sub-string-scale resolution. Phase Theory provides a minimum length via the measurement horizon μc , derived from admissibility without requiring extra dimensions or a landscape of vacua. String theory requires 10 or 11 spacetime dimensions; Phase Theory derives spacetime dimensionality from the relational substrate (Paper 3 [Hanks, 2026c]). String theory has not uniquely derived the Standard Model; Phase Theory claims (but has not yet fully demonstrated) that the Standard Model follows from the phase admissibility functional. Both frameworks agree that singularities are resolved and that a minimum length exists; they disagree on the mechanism and the accompanying theoretical complexity.

52.2 Phase Theory vs. Loop Quantum Gravity

LQG [Ashtekar & Rovelli, 1992] quantizes the gravitational field by promoting the holonomies and fluxes of the Ashtekar connection to operators, obtaining discrete spectra for area and volume. The minimum area eigenvalue is $A_{\min} = 4\pi \gamma \ell_P^2 \sqrt{3}/2$, where γ_{LQG} is the Barbero-Immirzi parameter. Phase Theory derives the minimum length from continuous admissibility, avoiding discretization and the associated violation of Lorentz invariance at the Planck scale. LQG has not derived the matter content of the Standard Model; Phase Theory has a systematic program for doing so via the structural freedom bound (§24). Both agree on singularity resolution; the Loop Quantum Cosmology Big Bounce is the Phase Theory analog of the minimum cosmological length.

52.3 Phase Theory vs. Causal Set Theory

Causal set theory [Bombelli et al., 1987] proposes that spacetime is fundamentally discrete, with events forming a partially ordered set. The discrete structure provides a natural cutoff for UV divergences. Phase Theory derives causal structure from the coherence horizon (§5): causality emerges from the finite propagation speed of coherent phase modes, not from a discrete underlying structure. Phase Theory recovers the causal set prediction that the cosmological constant is of order $1/N$ (where N is the number of causal events in the observable universe) from the vacuum noise floor (§20), but derives this from a continuous framework without requiring spacetime discreteness.

52.4 Phase Theory vs. Thermodynamic Bounds

The Bekenstein bound, Margolus-Levitin theorem, and Holevo bound are standard results in quantum information and thermodynamics. Phase Theory subsumes all of these as special cases (§16, §32, §37) and provides: (a) a unified origin for all three from the admissibility functional; (b) corrections in the high-curvature regime where standard derivations break down; (c) a derivation of the bounds without assuming quantum mechanics or thermodynamics as primitives. This places Phase Theory in the position of a more fundamental theory from which the thermodynamic bounds follow as consequences.

52.5 Phase Theory vs. It-from-Bit

Wheeler's "it from bit" program [Wheeler, 1990] proposes that physical reality emerges from information. Phase Theory agrees that information is fundamental but goes further: phase is more fundamental than information, because information requires distinguishable configurations, and distinguishability is derived from the admissibility functional. Information is a derived concept in Phase Theory; it is not the starting point. The holographic principle, emergent gravity (Verlinde, 2011), and AdS/CFT (Maldacena, 1997) all find natural accommodations in Phase Theory: the holographic principle follows from Theorem 15.7, entropic gravity from the relationship between phase coherence and gravitational curvature (§29), and AdS/CFT from the relational coherence structure of the phase substrate.

§53. Experimental Predictions

We enumerate eight classes of falsifiable predictions arising from Phase Theory's ultimate limits framework. For each, we specify the experimental protocol, expected signal size, and discriminating power relative to the Standard Model.

Prediction P1: Curvature-Bounded Hawking Spectrum.

Prediction: The thermal Hawking spectrum receives corrections of order $(E/E_{\text{Pl}})^2$ for photon energy E , deviating from exact thermality at near-Planckian temperatures. **Protocol:** Study near-extremal rotating black hole analogs in superfluid helium or Bose-Einstein condensates; measure the phonon emission spectrum with sub-percent precision. **Signal size:** For a BEC analog at $T_{\text{analog}} \sim 1$ nK, the Phase Theory correction is at the 10^{-6} level relative to the thermal spectrum. **Discriminating power:** Standard Hawking radiation predicts an exact Planck spectrum; any systematic deviation in the high-frequency tail distinguishes Phase Theory from Hawking's original calculation.

Prediction P2: Coherence Horizon Signatures in CMB.

Prediction: Non-Gaussianities in the CMB at angular scales corresponding to the cosmological coherence horizon $\ell_{\text{c}}^{\text{cosm}}$ deviate from slow-roll inflation predictions by the Phase Theory spectral correction $|\delta n_s| \sim H_0^2 / \kappa_{\text{max}}$. **Protocol:** High-resolution CMB polarization measurements (CMB-S4, LiteBIRD) with sensitivity to $f_{\text{NL}} \sim 0.1$. **Signal size:** $|\delta n_s| \sim 10^{-4}$, at the edge of CMB-S4 sensitivity. **Discriminating power:** Inflation models predict specific f_{NL} patterns; Phase Theory predicts a correlation between the non-Gaussianity amplitude and the coherence-horizon angular scale.

Prediction P3: Measurement Horizon Corrections to Atomic Spectra.

Prediction: Atomic transition frequencies receive corrections at the 10^{-20} level from the finite measurement horizon, detectable as a systematic deviation from QED predictions. **Protocol:** Compare 1S-2S hydrogen transition frequency with QED prediction at 17th-decimal-place precision using next-generation optical lattice clocks. **Signal size:** $|\delta\nu/\nu| \sim (\mu_c/a_0)^2 \sim 10^{-50}$, far below current technology but motivating long-term precision spectroscopy programs. **Discriminating power:** QED predicts no corrections at this level; Phase Theory predicts a universal sub-Planckian correction independent of the specific atomic system.

Prediction P4: Correlated Constant Drift.

Prediction: α_{em} , G, and c drift at correlated rates satisfying (15.55), with maximum rates $|\dot{X}/X| \leq H_0 \sim 10^{-18} \text{ yr}^{-1}$. **Protocol:** Multi-messenger astronomy combining: (a) spectroscopic measurement of $\dot{\alpha}_{em}$ via high-redshift quasar absorption spectra; (b) gravitational wave timing to measure $\dot{G}$ via binary pulsar orbital decay; (c) comparison of atomic clock frequencies to measure $\dot{c}/c$. **Signal size:** At $H_0 \sim 10^{-18} \text{ yr}^{-1}$, the drift of α_{em} over a 10-year campaign is $\sim 10^{-17}$, at the limit of NIST Sr clock comparisons. **Discriminating power:** The correlation ratio $(\dot{\alpha}_{em}/\alpha_{em}) / (\dot{G}/G) = \text{const}$ is a unique Phase Theory signature; Brans-Dicke and dilaton models predict different correlation structures.

Prediction P5: Maximum Particle Mass.

Prediction: No elementary particle with mass above $M_{\text{max}} \sim M_{\text{Pl}}/2 \approx 10^{19} \text{ GeV}/c^2$ can exist. **Protocol:** Any future collider operating at

sufficient center-of-mass energy. **Signal size:** No new particle production above $M_{\{max\}}$. **Discriminating power:** Standard Model does not predict a strict mass ceiling for composite or virtual particles; Phase Theory predicts a hard cutoff. Note that this prediction is currently inaccessible at existing colliders (LHC reach: ~ 14 TeV) but provides a well-defined target for post-LHC planning.

Prediction P6: Phase Noise Floor in Quantum Devices.

Prediction: All quantum amplifiers, regardless of design, exhibit a minimum noise floor of $|\sigma_{\{min\}}|^2 = \hbar / (\alpha |ell_c|^2 \tau_c)$, slightly exceeding the standard quantum limit. **Protocol:** Measure noise temperature of a series of state-of-the-art quantum amplifiers (HEMT, Josephson parametric amplifiers) at different physical scales $|ell_c|$ and compare with the Phase Theory prediction. **Signal size:** The Phase Theory excess noise is $\sim (|ell_c| / |\mu_c|)^{-2} \times T_{\{SQL\}}$ where $T_{\{SQL\}}$ is the standard quantum limit; at $|ell_c| \sim 1$ cm, this is approximately $10^{-66} T_{\{SQL\}}$ — undetectable currently but setting an in-principle limit. **Discriminating power:** Standard quantum mechanics predicts the SQL as the absolute minimum; Phase Theory predicts the SQL is not achievable, with a correction of order $(|\mu_c|/|ell_c|)^2$.

Prediction P7: Refractive Gravity Deviations Near k_{max} .

Prediction: High-energy photons traveling near compact objects (neutron stars, near-extremal black holes) experience an energy-dependent deflection angle deviating from GR by $|\delta\theta|/\theta_{\{GR\}} \sim (E/E_{\{Pl\}})^2$. **Protocol:** Multi-wavelength VLBI observations of photons with energies from radio to gamma-ray near neutron stars; compare deflection angles. **Signal size:** For TeV-energy gamma rays ($E \sim 10^{12}$ eV) and $E_{\{Pl\}} \sim 10^{28}$ eV, the correction is $\sim 10^{-32}$ — far too small for current technology, but motivating next-generation gamma-ray interferometry programs.

Discriminating power: GR predicts no energy-dependent deflection (all photons deflected equally); any energy dependence is a Lorentz invariance violation, which Phase Theory predicts at order $(E/E_{Pl})^2$ specifically.

Prediction P8: Entanglement Horizon.

Prediction: Quantum entanglement cannot be maintained over spatial separations exceeding the relational coherence horizon $|ell_c(r)| \sim |ell_c^{cosm}| \sim c/H_0$. At intermediate baselines of order 10^4 – 10^6 km, a systematic reduction in entanglement fidelity beyond what decoherence models predict should be observable. **Protocol:** Satellite-based quantum communication experiments (extending existing ground-to-satellite protocols) with baselines of 10^4 km and below; measure Bell inequality violation as a function of baseline and compare with decoherence-corrected QM predictions. **Signal size:** The Phase Theory correction to entanglement fidelity scales as $(b/|ell_c(r)|)^2$ for baseline b ; at $b = 10^4$ km and $|ell_c(r)| \sim 10^{26}$ m (Hubble scale), this is $\sim 10^{-38}$ — undetectable currently, but Phase Theory predicts non-trivial structure at sub-Hubble relational scales if $|ell_c(r)| \ll c/H_0$ (Open Problem O5). **Discriminating power:** Standard QM predicts no fundamental limit on entanglement range; any systematic drop in fidelity not explained by environmental decoherence would be a direct test of Phase Theory.

§54. Open Problems

We enumerate the major open problems in the Phase Theory ultimate limits framework.

O1: Uniqueness of the admissibility functional. Is the admissibility functional $I[\Phi]$ uniquely determined by the Phase Theory axioms, or is there a family of

consistent admissibility functionals? The structure of L_{admiss} in (15.2) was motivated but not uniquely derived from first principles. A more rigorous derivation (sketched in Appendix B) is needed to determine whether $(\alpha, \beta, \gamma, \lambda)$ are uniquely fixed or span a moduli space. If a family exists, the size of the family determines the "ambiguity" in Phase Theory predictions.

O2: Dynamical computation of κ_{max} , ρ_r^{max} , and μ_c . The limits derived in this paper are expressed in terms of the admissibility parameters $(\alpha, \beta, \gamma, \lambda)$. Computing their numerical values from first principles requires solving the full admissibility functional nonperturbatively — a problem analogous to computing the QCD scale Λ_{QCD} from the QCD Lagrangian. Current numerical estimates (Appendix D) use phenomenological matching; a first-principles calculation is a major outstanding challenge.

O3: Phase Theory at finite temperature. At nonzero temperature, thermal fluctuations $|\delta\Phi_{therm}| \sim k_B T / \hbar$ are superimposed on quantum phase fluctuations. How do coherence horizons depend on temperature? Does $\ell_c^{(s)}(T)$ decrease monotonically with T (as for correlation lengths in condensed matter), and how does this affect the stability of physical constants? The finite-temperature Phase Theory framework has not been fully developed.

O4: Gravitational coherence. The gravitational field in Phase Theory emerges from the curvature coupling of $I/|\Phi|$. Does the gravitational field itself have a coherence horizon, and if so, what are its implications? Specifically: does the graviton (the quantum of the gravitational field in Phase Theory) have a coherence horizon shorter than the cosmological scale, implying quantum gravity decoherence? If so, this would predict a loss of quantum superposition for massive objects above a mass threshold set by the gravitational coherence horizon — a testable prediction related to Penrose's gravitational decoherence hypothesis.

O5: Relational substrate geometry. The intrinsic geometry of $|\Omega$ — its dimension, topology, and metric — has not been fully specified beyond "n-dimensional Riemannian manifold." The relational coherence horizon $\ell_c^{(r)}$ depends sensitively on this geometry. A derivation of the relational substrate

geometry from Phase Theory axioms would provide a definitive prediction for the maximum entanglement range (Prediction P8).

O6: Phase Theory renormalization. The admissibility functional parameters (α , β , γ , λ) are scale-dependent in the same sense as coupling constants in quantum field theory. The Phase Theory "renormalization group" (the running of these parameters with coherence scale) has not been computed. Understanding this running is necessary for connecting Planck-scale Phase Theory to low-energy physics predictions.

O7: Connection to quantum information theory. Phase Theory bounds (§31–§35) bear strong resemblance to quantum channel capacity bounds (Holevo, Shannon-Hartley, quantum channel capacity). Can Phase Theory limits be expressed as quantum channel capacities in a rigorous information-theoretic framework? This would connect Phase Theory to the operational quantum information theory framework and provide new tools for computing bounds.

O8: Experimental accessibility. Of the eight predictions in §53, which are achievable within the next decade? A careful analysis suggests: P4 (correlated constant drift) is the most accessible, with next-generation atomic clocks and pulsar timing arrays potentially reaching the required precision by 2035; P1 (Hawking spectrum) requires improved BEC analog experiments achievable within a decade; P2 (CMB non-Gaussianities) will be testable with CMB-S4 (expected 2030). P3, P5, P6, P7, and P8 require technology beyond the current decade. Priority experimental investment should focus on P4 and P1.

§55. Conclusion

We have derived, from a single principle — the admissibility condition $I[\Phi] \geq 0$ — the complete set of ultimate physical limits governing phase coherence, information density, curvature, computation, observation, and constant variation. The central result of this paper may be stated as the following master theorem:

Master Theorem (Ultimate Phase Limits).

Let Φ be any physically realized phase configuration. Then Φ lies in the existence regime E , satisfying simultaneously: (a) $\chi[\Phi] \geq \epsilonpsilon_c$ (anti-fragmentation), (b) MPC (anti-freezing), (c) $\kappa[\Phi] \leq \kappa_{\max}$ (maximum curvature), and (d) $\rho_I[\Phi] \leq \rho_I^{\max}$ (maximum information density). These four bounds together guarantee: singularity prevention, finite complexity, stable universal constants, bounded computation and observation, and the existence of observers without fine-tuning or anthropic selection.

The eight main results of this paper are:

(R1) Coherence horizons exist and are finite (Theorem 15.1). Every non-trivial admissible phase configuration has a finite coherence horizon $\ell_c < \infty$, providing the phase-theoretic basis for locality, causality, finite interaction ranges, and the emergence of spacetime structure.

(R2) Phase curvature is bounded, forbidding singularities (Proposition 15.3, Theorem 15.6). The maximum admissible curvature $\kappa_{\max} = \gamma/(2\beta) + \lambda T_{\max}/\beta$ is finite and corresponds to the Planck curvature. No physical observable diverges in an admissible configuration; classical singularities are structurally excluded.

(R3) Information density is bounded, forbidding infinite complexity (Theorem 15.10, §15). The maximum information density $\rho_I^{\max} \leq \kappa_{\max}^{n/2}/(2\pi)^n$ is finite, providing a universal UV cutoff without lattice discreteness and reproducing the Bekenstein-Hawking and holographic bounds as special cases.

(R4) Reality occupies the unique stability regime E (Proposition 15.7, Theorem 15.11). The intersection of the anti-fragmentation bound, anti-freezing condition, curvature bound, and information density bound defines the existence regime E , which is non-empty, open, connected, and invariant under admissible

phase evolution. Physical reality neither fragments nor freezes; it persists in E indefinitely.

(R5) Universal constants emerge from admissibility parameters (§26-§30).

The speed of light, Planck constant, gravitational constant, and fine structure constant are not fundamental free parameters but emergent conversion factors derived from $(\alpha, \beta, \gamma, \lambda)$. Their values are fixed (up to cosmological drift) by the admissibility functional.

(R6) Computation, observation, and error correction are bounded (§31-§39).

The maximum computational rate, the minimum detectable signal, the error correction depth, and the unavoidable noise floor are all derivable from the admissibility bounds. These Phase Theory bounds subsume the Margolus-Levitin theorem, standard quantum limit, Bekenstein bound, and Holevo bound as special cases with Planck-scale corrections.

(R7) Constants drift slowly under admissibility (§40-§41). Emergent constants can vary at most at the Hubble rate $H_0 \sim 10^{-18} \text{ yr}^{-1}$, with correlated drift across all constants. This provides a distinctive and falsifiable observational signature accessible with next-generation instruments.

(R8) Observers are generic, not anthropically special (Theorems 15.18-15.20). Every admissible configuration in E supports complexity far exceeding the minimum required for observers. The fine-tuning problem is dissolved: constants are not fine-tuned because they are not free parameters. Anthropic selection is unnecessary because observers are generic consequences of admissibility.

The philosophical implication is profound and deserves emphasis. Phase Theory demonstrates that the stability, finitude, and richness of physical reality are not contingent features that require external explanation, anthropic selection, or laws imposed from outside. They are necessary consequences of the internal structure of admissibility. Reality does not need external rules to remain stable, bounded, and complex. The admissibility functional is simultaneously the definition of physical existence and the guarantor of its persistence. Reality is, in the deepest sense, *self-bounded by phase*.

Future directions in Phase Theory include the resolution of the eight open problems enumerated in §54, the development of a full nonperturbative treatment of $I[|\Phi|]$, and the experimental program outlined in §53. The Phase Theory series continues in Paper 16 with a detailed treatment of the relational substrate geometry and its implications for the maximum entanglement range.

Appendix A: Mathematical Notation and Conventions

We provide a complete glossary of all symbols used in this paper.

Symbol	Definition	First Used
Φ	Phase field: $\Phi: \Omega \rightarrow S^1$	§1
Ω	Relational substrate (measure space)	§1
$I[\Phi]$	Admissibility functional	§1
L_{admiss}	Admissibility Lagrangian density	§2
α	Gradient penalty coefficient in L_{admiss}	§2
β	Curvature penalty coefficient in L_{admiss}	§2
γ	Coherence reward coefficient in L_{admiss}	§2
λ	Topological term coefficient in L_{admiss}	§2
$\kappa[\Phi](x)$	Phase curvature: $ \nabla^2 \Phi ^2$	§2
K_{max}	Maximum admissible phase curvature	§11
$C(\Phi)$	Coherence term: $ f e^{i\Phi} d\Omega ^2 / \Omega ^2$	§2
$T(\Phi)$	Topological winding term	§2
$\chi[\Phi; V]$	Phase coherence measure for region V	§3
ϵ_c	Minimum admissible coherence threshold	§6
$G(x, x')$	Two-point phase	§6

Symbol	Definition	First Used
	correlation function	
$\ell_c[\Phi]$	Coherence horizon of configuration Φ	§6
$\ell_c^{(s)}$	Spatial coherence horizon	§7
$\ell_c^{(r)}$	Relational coherence horizon	§9
ℓ_c^{cosm}	Cosmological coherence horizon	§10
τ_c	Temporal coherence horizon	§8
v_Φ	Phase propagation speed (= c in emergent limit)	§10
ρ_Φ	Phase information density $ \nabla\Phi ^2$	§7
ρ_I^{max}	Maximum information density per coherence volume	§14
V_c	Coherence volume ℓ_c^n	§14
δI_{min}	Minimum distinguishable action increment (= $\hbar$)	§14
σ_{max}^2	Maximum admissible phase noise variance	§20
σ_{min}^2	Minimum (unavoidable) noise floor	§34
μ_c	Measurement horizon (minimum length)	§39
P	Physical regime: $\{\Phi : I[\Phi] \geq 0\}$	§25
E	Existence regime (intersection of four stability conditions)	§43
D[V]	Structural freedom of coherence region V	§24
N_c	Distinguishable configurations per coherence cell	§24
N_{top}	Total topological charge	§4
R	Computational rate	§32

Symbol	Definition	First Used
	(transitions per unit time)	
D	Error correction depth	§33
G_{DA}	Distinguishability amplification gain	§37
$C[\Phi]$	Complexity of phase configuration	§45
S_{holo}	Phase holographic entropy	§17
c_{Φ}	Phase propagation speed $(\alpha/\kappa)^{1/2} = c$	§27
k_{max}	Maximum admissible wavenumber $(\alpha/\beta)^{1/2}$	§27
Φ_0	Phase vacuum (unique maximum-coherence admissible configuration)	§19
$\delta^2 I$	Second variation of admissibility functional	§19
MPC	Marginal Permissivity Condition	§23
I_c	Critical admissibility threshold ($= 0$)	§2
$d_{\Omega}(x, x')$	Relational distance on Ω	§9

Conventions. We work in units where the admissibility functional has dimensions of action. Physical units are restored by identifying $|\delta I_{\min}| = \hbar$ (§28), $v_{|\Phi|} = c$ (§27), and $G = |\alpha|^2/(16\pi\kappa^2)$ (§29). Metric signature: $(+, -, -, -)$ for the emergent Lorentzian spacetime. Greek indices μ, ν, ρ, σ run over spacetime dimensions 0–3. Latin indices i, j, k run over spatial dimensions 1–3. Repeated indices are summed. $|/\Omega|$ denotes the relational measure of $|\Omega|$. $|\partial_{\mu}|$ denotes partial derivative with respect to the μ -th relational coordinate. $|\nabla^2 = g^{\{\mu\nu\}}|\nabla_{\mu}|\nabla_{\nu}|$ is the Laplace-Beltrami operator on $|\Omega|$. $|square$ denotes QED (end of proof). All logarithms are natural unless base 2 is explicitly indicated (for information-theoretic quantities in bits).

Appendix B: Derivation of the Admissibility Functional from First Principles

The admissibility functional $I[\Phi]$ is derived from the five axioms of Phase Theory stated in Paper 1 [Hanks, 2026a]. We sketch the derivation here with cross-references.

Axiom A1 (Phase Ontology): The sole fundamental constituent of physical reality is the phase field $\Phi: \Omega \rightarrow S^1$. All physical structures are configurations of Φ .

Axiom A2 (Phase Symmetry): The physical description is invariant under global phase rotations $\Phi \rightarrow \Phi + \phi_0$ (constant shift) and local phase gauge transformations coupled to gauge fields (Paper 5 [Hanks, 2026e]).

Axiom A3 (Admissibility): A phase configuration is physically realizable if and only if it is admissible: $I[\Phi] \geq 0$.

Axiom A4 (Global Consistency): The admissibility condition must be satisfied globally, not just locally. Physical configurations must respect the integral constraint over all of Ω .

Axiom A5 (Minimality): The admissibility functional is the simplest functional of Φ and its derivatives up to second order that: (a) is positive for the vacuum $\Phi = \text{const}$, (b) is invariant under Axiom A2 symmetries, (c) penalizes incoherence, and (d) penalizes high curvature.

From Axiom A5, the most general functional satisfying conditions (a)–(d) and containing at most second-order derivatives is:

$$(B.1) \quad I[\Phi] = \int_{\Omega} [f_1(C(\Phi)) + f_2(|\nabla\Phi|^2) + f_3(|\nabla^2\Phi|^2) + f_4(T(\Phi))] d^n\Omega,$$

where f_1, f_2, f_3, f_4 are functions constrained by (a)–(d). Condition (a) requires $f_1(1) > 0$ and $f_2(0) = f_3(0) = 0$. Condition (c) requires $f_1'(C) > 0$ (coherence reward). Condition (d) requires $f_3'(\kappa) < 0$ (curvature penalty). The simplest choice consistent with all conditions is the linear-in-each-variable form of (15.2).

Non-linear forms ($f_1(C) = \gamma C^2$, etc.) would give qualitatively similar physics but modified quantitative predictions; the linear form is selected by Axiom A5 (minimality/Occam). This completes the derivation sketch. A rigorous derivation using functional analysis is in preparation for Paper 16.

Appendix C: Relation Between Phase Theory Bounds and Thermodynamic Limits

Phase Theory bounds and thermodynamic limits share deep structural connections. We systematically compare them.

Second Law and Anti-Fragmentation. The thermodynamic second law states that entropy never decreases in a closed system. The Anti-Fragmentation Theorem (§18) states that coherence never falls below ϵ_c under admissible evolution. These are related but not identical: entropy is a measure of incoherence (high entropy = low coherence), so the second law implies entropy increases while the Anti-Fragmentation Theorem bounds how high entropy can go. The Phase Theory version is: entropy is bounded above by $S_{\max} = k_B \log(\rho_I^{\max} V)$, not unbounded as in standard thermodynamics. This represents an important correction: Phase Theory is not compatible with heat death in the traditional sense, because the information density bound prevents entropy from reaching its classical maximum.

Maximum Entropy Principle and Existence Regime. The maximum entropy principle states that a thermodynamic system at equilibrium has the configuration that maximizes entropy subject to constraints. In Phase Theory, the "equilibrium" configuration maximizes information density subject to admissibility: it is the configuration at the information boundary of E (§43). This maximum-information configuration is not the same as the classical maximum-entropy state, because Phase Theory's information density is bounded while classical entropy is not (for continuous systems).

Fluctuation Theorems and Phase Noise. The Crooks fluctuation theorem and Jarzynski equality relate the work performed in nonequilibrium processes to

equilibrium free energy differences. In Phase Theory, these correspond to relations between the admissibility change $|\Delta I|$ and the phase noise amplitude $|\sigma|^2$. Specifically, the fluctuation-dissipation relation from Phase Theory is $|\sigma|^2 = 2k_B T |\alpha| / (|\alpha|^2 |\ell_c|^2)$ in thermal equilibrium, reducing to the standard result $|\sigma|^2 = 2k_B T / (|\alpha| |\ell_c|^2)$ in the classical limit. The Phase Theory corrections become relevant at temperatures $k_B T \sim \hbar v_\Phi / |\ell_c|$ (the quantum-classical crossover), providing modifications to the fluctuation theorems near the quantum limit.

Thermodynamic Limit and Coherence Horizon. In statistical mechanics, the thermodynamic limit is the limit of infinite system size with fixed density. In Phase Theory, the thermodynamic limit corresponds to the limit $V \rightarrow \infty$ with fixed $|\ell_c|$. The coherence horizon ensures that this limit is well-defined: even for infinite V , the system decomposes into independent coherence volumes, each with finite properties. The Phase Theory thermodynamic limit is therefore automatically well-defined, unlike the classical case where long-range correlations can cause the thermodynamic limit to fail (e.g., gravitational systems).

Appendix D: Numerical Estimates of Phase Theory Constants

We compute numerical estimates for the key Phase Theory constants in SI units, using the measured values $c = 2.998 \times 10^8$ m/s, $\hbar = 1.055 \times 10^{-34}$ J·s, and $G = 6.674 \times 10^{-11}$ N·m²/kg².

Quantity	Phase Theory Expression	Numerical Estimate	Comparison / Notes
Measurement horizon μ_c	$(\hbar/\gamma)^{1/3}$	$\sim 1.6 \times 10^{-35}$ m	$= \ell_{\text{Pl}}$ (Planck length)
Maximum curvature κ_{max}	$\kappa_{\text{max}} = c^7/(\hbar G^2)$	$\sim 5.2 \times 10^{93}$ m ⁻⁴	$=$ Planck curvature K_{Pl}
Info. density $\rho_{\text{I}}^{\text{max}}$	$\kappa_{\text{max}}^{3/2}/(2\pi)^3$	$\sim 10^{141}$ bits/m ³	~ 1 bit per ℓ_{Pl}^3
Coherence coefficient γ	$\hbar/\mu_c^3$	$\sim 2.5 \times 10^{69}$ J/m ³	$=$ Planck energy density
Gradient	$c^2/\kappa_{\text{max}}^{1/2}$	$\sim 4.2 \times 10^{-53}$ m ² /s ²	Sets phase

Quantity	Phase Theory Expression	Numerical Estimate	Comparison / Notes
coefficient α			propagation speed = c
Curvature coefficient β	$G/(4\pi) = \alpha^2/(4\kappa^2)$	$\sim 5.3 \times 10^{-12} \text{ N}\cdot\text{m}^2/\text{rad}^2$	Sets $G = 6.674 \times 10^{-11}$
Spatial coherence $\ell_c^{(s)}$	$(\kappa/\alpha)^{1/2}$	Scale-dependent: from ℓ_{Pl} to c/H_0	Mode-dependent; see §7
Cosm. coherence ℓ_c^{cosm}	c/H_0	$\sim 1.3 \times 10^{26} \text{ m}$	= Hubble radius
Temporal coherence τ_c	$\ell_c^{(s)}/v_\Phi$	Scale-dependent: from t_{Pl} to H_0^{-1}	Mode-dependent; see §8
Vacuum energy ρ_Λ	$\hbar H_0^4/c^3$	$\sim 5 \times 10^{-27} \text{ kg/m}^3$	Obs. value $\sim 6.9 \times 10^{-27} \text{ kg/m}^3$
Max. particle mass M_{max}	$M_{\text{Pl}}/2$	$\sim 1.1 \times 10^{19} \text{ GeV}/c^2$	= Planck mass / 2
Max. constant drift	H_0	$\sim 2.3 \times 10^{-18} \text{ yr}^{-1}$	$\sim 70 \text{ km/s/Mpc}$

The remarkable feature of these estimates is the self-consistency of the framework: starting from c , $\hbar$, G as empirical inputs, Phase Theory reproduces the Planck scale quantities exactly (by construction of the identification in §27–§29), the cosmological constant to within a factor of ~ 1.4 (without any free parameters adjusted for this purpose), and the qualitative structure of all other bounds. The numerical estimates are to be regarded as order-of-magnitude checks, not precision calculations; a full nonperturbative treatment is required for exact values (Open Problem O2).

References

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- [1] Hanks, M. (2026a). *Phase Theory: Axioms and Foundations*. Phase Theory Series, Paper 1. Preprint.
- [2] Hanks, M. (2026b). *The Particle Spectrum from Phase Admissibility*. Phase Theory Series, Paper 2. Preprint.

- [3] Hanks, M. (2026c). *Emergent Spacetime from Phase Propagation*. Phase Theory Series, Paper 3. Preprint.
- [4] Hanks, M. (2026d). *Phase-Theoretic Gravity and the Emergence of General Relativity*. Phase Theory Series, Paper 4. Preprint.
- [5] Hanks, M. (2026e). *Gauge Theory from Phase Symmetry*. Phase Theory Series, Paper 5. Preprint.
- [6] Hanks, M. (2026f). *Quantum Entanglement as Relational Phase Coherence*. Phase Theory Series, Paper 6. Preprint.
- [7] Hanks, M. (2026g). *Topological Defects and Phase Sectors*. Phase Theory Series, Paper 7. Preprint.
- [8] Hanks, M. (2026h). *Quark Confinement from Phase Admissibility*. Phase Theory Series, Paper 8. Preprint.
- [9] Hanks, M. (2026i). *Phase Information Theory: Entropy, Distinguishability, and Capacity*. Phase Theory Series, Paper 12. Preprint.
- [10] Hanks, M. (2026m). *Phase-Theoretic Black Holes*. Phase Theory Series, Paper 13. Preprint.
- [11] Bekenstein, J. D. (1973). Black holes and entropy. *Physical Review D*, 7(8), 2333–2346.
- [12] Hawking, S. W. (1975). Particle creation by black holes. *Communications in Mathematical Physics*, 43(3), 199–220.
- [13] Penrose, R. (1965). Gravitational collapse and space-time singularities. *Physical Review Letters*, 14(3), 57–59.
- [14] Margolus, N., & Levitin, L. B. (1998). The maximum speed of dynamical evolution. *Physica D: Nonlinear Phenomena*, 120(1–2), 188–195.
- [15] Lloyd, S. (2000). Ultimate physical limits to computation. *Nature*, 406(6799), 1047–1054.
- [16] Planck, M. (1899). Über irreversible Strahlungsvorgänge. *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften*, 5, 440–480.
- [17] Holzhey, C., Larsen, F., & Wilczek, F. (1994). Geometric and renormalized entropy in conformal field theory. *Nuclear Physics B*, 424(3), 443–467.

- [18] Ryu, S., & Takayanagi, T. (2006). Holographic derivation of entanglement entropy from the anti-de Sitter/conformal field theory correspondence. *Physical Review Letters*, 96(18), 181602.
- [19] Wheeler, J. A. (1990). Information, physics, quantum: The search for links. In W. H. Zurek (Ed.), *Complexity, Entropy, and the Physics of Information*. Addison-Wesley.
- [20] Deutsch, D. (1985). Quantum theory, the Church-Turing principle and the universal quantum computer. *Proceedings of the Royal Society of London A*, 400(1818), 97-117.
- [21] Susskind, L. (1995). The world as a hologram. *Journal of Mathematical Physics*, 36(11), 6377-6396.
- [22] Ashtekar, A., & Rovelli, C. (1992). A weave for kinematical excitations of gravity. *Classical and Quantum Gravity*, 9(5), 1121-1150.
- [23] Bombelli, L., Lee, J., Meyer, D., & Sorkin, R. D. (1987). Space-time as a causal set. *Physical Review Letters*, 59(5), 521-524.
- [24] Verlinde, E. (2011). On the origin of gravity and the laws of Newton. *Journal of High Energy Physics*, 2011(4), 029.
- [25] Maldacena, J. (1997). The large N limit of superconformal field theories and supergravity. *International Journal of Theoretical Physics*, 38(4), 1113-1133.
- [26] Penrose, R. (2004). *The Road to Reality: A Complete Guide to the Laws of the Universe*. Jonathan Cape, London.
- [27] Bekenstein, J. D. (1981). Universal upper bound on the entropy-to-energy ratio for bounded systems. *Physical Review D*, 23(2), 287-298.
- [28] Penrose, R. (1971). Angular momentum: an approach to combinatorial spacetime. In T. Bastin (Ed.), *Quantum Theory and Beyond*. Cambridge University Press.
- [29] Rovelli, C., & Smolin, L. (1995). Discreteness of area and volume in quantum gravity. *Nuclear Physics B*, 442(3), 593-619.
- [30] Page, D. N. (1993). Information in black hole radiation. *Physical Review Letters*, 71(23), 3743-3746.

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January 9, 2026 | Preprint, Under Review | Independent Researcher

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